Party Lines or Voter Preferences? Explaining Political Realignment

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Abstract

This paper estimates a political equilibrium model to disentangle demand factors (voters) from supply factors (politicians) in shaping political outcomes, focusing on the recent realignment of blue-collar voters away from left-wing parties. I jointly evaluate the impact of changes in voter preferences and voter demographics (demand side) and party positions and party discipline (supply side) on voters' partisan realignment as observed in U.S. House elections between 2000 and 2020. To measure candidate ideological positioning, I estimate a multimodal text-and-survey model using campaign websites. To estimate voter preferences, I build a new panel of precinct-level election results (N=1.6 million), which allows me to exploit congressional district border discontinuities for identification. The paper ultimately identifies parties' stronger polarization on cultural issues compared to economic issues as the main driver of voters' partisan realignment. In contrast, shifts in voter preferences—particularly the increasing support among blue-collar voters for progressive economic policies—have mitigated their defection from the Democratic Party. Absent these demand-side changes, partisan realignment would have been even more pronounced.

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1 Introduction

While traditionally favored by the majority of less-educated voters, left-wing parties in Western democracies have seen their base shift significantly toward a more educated electorate (Kitschelt and Rehm, 2019; Gethin et al., 2022). This trend is widespread, but the pace and intensity of this realignment in the United States stand out as particularly striking. Over the past 20 years, the proportion of voters with a high school diploma or less supporting the Democratic Party has dropped by more than 10 percentage points, while the share of college graduates supporting the Democratic Party has increased by a similar margin.[1] This realignment has not only altered voting patterns but also led to profound transformations in political polarization and policymaking, reshaping the structure of party competition in the United States.

The literature is divided between demand-side and supply-side explanations of political realignment. On the one hand, some argue that voters' shifting preferences—such as an increasing focus on cultural issues—are the primary driver behind their realignment (Inglehart, 1997; Enke et al., 2021; Danieli et al., 2022; Noy and Rao, 2025).[2] On the other hand, supply-side explanations suggest that the key factor is parties' changing positions (Rennwald and Evans, 2014; Choi et al., 2024; Kuziemko et al., 2026). No consensus has emerged, largely due to the challenge of jointly assessing changes in voter behavior and party strategies within a single framework.

This paper directly addresses this gap by offering a unified empirical framework that integrates both demand and supply factors to assess changes in political equilibrium outcomes. Specifically, the main empirical goal is to estimate the contributions of changes in voter preferences and voter demographics (demand-side changes) and changes in party leadership positions and party discipline (supply-side changes) to this recent voter realignment episode. The main takeaway is that supply-side factors accounted for most of the political realignment among less-educated voters, while demand-side factors pushed in the opposite direction. Absent changes in voter preferences, political realignment by education would have been even more pronounced.

Understanding the drivers of political realignment requires making progress on longstanding questions in political economy: How does the interaction between voters and parties determine political outcomes in equilibrium? How do voters respond to policies offered by their local candidates? And are candidates catering their positions to their constituents, or only following their party line? Answering these questions has proven difficult due to a series of independent empirical challenges.

The first challenge concerns the measurement of candidate positions themselves; grappling with the multidimensionality of ideological positioning and comparing these dimensions across multiple elections

[1] This coalition change did not go unnoticed by party leaders. In 2016, Chuck Schumer stated that "for every blue-collar Democrat we lose in western Pennsylvania, we will pick up two moderate Republicans in the suburbs of Philadelphia," suggesting that some Democratic elites viewed this realignment as part of a broader electoral strategy.

[2] Noy and Rao (2025) provide evidence that cable news indirectly contributed to the rise of cultural politics by shifting voters' issue priorities.

is particularly difficult (Poole and Rosenthal, 2011; Bateman and Lapinski, 2016). Additionally, previous studies have typically been limited to measures available only for incumbents, as obtaining reliable information on the positions of challengers has often been more complex. Second, to recover the distribution of voter preferences, it is necessary to observe multiple pairs of candidates, each offering their own platform, along with variation in voter choices for any given choice set (Berry and Haile, 2021). Third, identifying exogenous shifters for both demand and supply variation is essential to circumvent possible endogeneity concerns in the estimation of voters' and candidates' preferences (Berry and Haile, 2024). Finally, individual voting decisions are seldom observed and relying on aggregate voting data makes it difficult to recover individual preferences (King, 1997).

This paper makes progress on these issues in several ways. To begin with, I recover the position of all candidates (incumbents and challengers) in Congressional races on both cultural and economic dimensions by training a multimodal text-and-survey model based on candidate websites.[3] These dimensions are both interpretable and comparable over time thanks to candidate survey questions asked repeatedly. Next, to measure demand, I construct a new panel dataset of precinct-level election results in Congressional races between 2000 and 2020 to capture extremely granular, within-district variation in voter preferences (175,000 precincts per year). The granularity of the election data is key—not only for estimating the rich heterogeneity in voter preferences—but also for exploiting congressional district border discontinuities to identify these preferences. All in all, I estimate a political equilibrium model of U.S. House elections, quantifying the influence of supply-side factors, such as party leadership positions and party discipline, as well as demand-side factors, such as voter ideological preferences, and voter demographics.

The key finding is that voter realignment by education observed in the U.S. between 2000 and 2020 is primarily driven by supply-side factors. This conclusion rests on three main points. First, less-educated voters favor conservative cultural policies but progressive economic policies, with their preference for the latter increasing over time. Second, despite this shift in economic preferences, party polarization has increased much more sharply on cultural issues than on economic ones. Taken together, these points indicate that the larger polarization on cultural issues has pushed less-educated voters away from supporting Democratic candidates, even though their preference for progressive economic policies has grown. These changes in voter preferences, however, led less-educated voters to support Democratic candidates more. In other words, absent these demand-side changes, less-educated voters would have shifted even more toward Republican candidates, further deepening political realignment. Below, I outline the empirical framework and intermediate results that support this conclusion, drawing on both reduced-form evidence and structural estimation.

I begin by documenting descriptive changes in voting behavior over the sample period. The precinct-

[3] In Section 3, I provide a detailed explanation of the classification of each topic into cultural and economic dimensions. Economic topics typically include taxes, healthcare, welfare, trade policy, and labor relations, while cultural topics encompass crime, gun regulations, reproductive rights, education, LGBTQ rights, affirmative action, and environmental issues.

level panel data provide fine-grained variation,[4] allowing me to identify and compare the demographic variables along which this voter realignment has occurred. I show that realignment along educational lines has dominated realignment along any other demographic line, such as race, occupation, or union affiliation. Regarding candidate positioning, while party polarization is well-documented,[5] combining survey data with natural language processing applied to candidate websites shows that polarization on cultural issues has risen much more sharply than on economic ones, with two-thirds of this widening gap driven by the Democratic Party. This analysis applies to both winning and losing candidates.

In the next stage of the analysis, I estimate a political equilibrium model that captures two key aspects of the electoral process. First, candidates compete by selecting positions in the ideological space, aiming to maximize their chances of election while minimizing the ideological distance between their chosen positions and those of their party leadership. The weight given to party leadership reflects the strength of party discipline. Second, heterogeneous voters choose the candidate whose platform and characteristics best align with their preferences.

To recover voter preferences over candidate platforms, I estimate a structural model of voter behavior using fine-grained variation in election results from precinct data. I also report reduced-form estimates that exploit the same border variation, providing a direct counterpart to the structural demand estimates. Since candidate positions depend on unobservables affecting demand, I address their endogeneity by matching contiguous precincts across congressional district borders within each state, following the approach used for counties in Spenkuch and Toniatti (2018). Since candidates set strategies at the district level and each individual precinct (about 400 per district) has a negligible impact from the candidate perspective, differences in candidate positions between contiguous precincts across district borders can be treated as random, especially after accounting for time-invariant precinct characteristics. While unobservables cannot be directly tested, I provide evidence that, conditional on fixed effects, candidate positions are uncorrelated with a wide range of precinct-level demographics.

I find that more-educated voters favor progressive cultural positions, but not progressive economic ones. Less-educated voters, by contrast, have increasingly favored progressive economic positions, while the educational gradient on cultural issues has remained relatively stable.

Further, I document a rise in party discipline in both parties. Using estimates of the distribution of ideological preferences in each electoral precinct, I recover supply-side parameters measuring the weight candidates place on winning elections relative to aligning with the party when choosing the positions they offer to voters. I show that party discipline has increased sharply over time: House candidates' ability to adapt their positions to their constituents has been substantially reduced, leading to greater uniformity in candidate positions across districts.

Finally, combining the demand- and supply-side estimates, I solve the model under several counter-

[4]Each precinct contains on average 1,000 registered voters, that is, about 50 times more granular than a county and 400 times more granular than a congressional district.

[5]See e.g., McCarty et al. (2016), Gentzkow et al. (2019) for elected representatives, and Bonica (2013) for candidates.

factual scenarios, recovering the corresponding equilibrium candidate positions and vote shares. This exercise leads to the paper's key result: when assessing changes in voter choices, I show that virtually all of the explained political realignment is driven by supply-side factors, particularly the rising polarization between the two parties on cultural issues compared to economic issues.

In contrast, changes in voter preferences have been pushing in the opposite direction: the growing preference for progressive economic policies among less-educated voters has mitigated the extent of realignment. In other words, while less-educated voters have increasingly supported progressive economic policies, traditionally offered by the Democratic Party, the parties have become more polarized on cultural issues, pushing less-educated voters toward the Republican Party. These findings resonate with theories suggesting that cultural polarization has weakened democracy's ability to address rising inequality (Bonica et al., 2013; Roemer, 1998; Hacker and Pierson, 2020).

Related Literature This paper makes contributions to several strands of the literature in economics and political science.

First, this paper contributes to the literature estimating structural models of voter preferences, particularly papers that recover preferences from aggregate election results (Coate and Conlin, 2004; Rekkas, 2007; Strömberg, 2008; Gordon and Hartmann, 2013; Sieg and Yoon, 2017; Ujhelyi et al., 2021; Kawai and Sunada, 2022; Iaryczower et al., 2022; Cox, 2023; Cox and Shapiro, 2024; Berry et al., 2024; Magesan et al., 2026). A unique feature of this paper is the use of election results at a much more granular level than where supply decisions are made. The granularity of the election data is crucial—not only for estimating the rich heterogeneity in voter preferences—but also for exploiting congressional district border discontinuities to address endogeneity in candidate ideology. To the best of my knowledge, this approach offers the first estimates of individual voter preferences over candidate ideology in the United States.

Many articles have also inferred voter preferences from correlations between voters' reported electoral choices in survey data and party or candidate positions (Adams et al., 2004; Elff, 2009; Evans and Tilley, 2012; Krasa and Polborn, 2014; Danieli et al., 2022). In particular, many papers rely on issue positioning from the Comparative Manifesto Project. This paper complements this literature by exploiting revealed rather than stated preferences and using plausibly exogenous variation in candidate positions to recover voter preferences. The framework developed in this paper leverages the multiplicity of House candidates and granular precinct-level data, exploiting district border discontinuities to address the endogeneity of candidate positions.

A key strength of this paper is the integration of demand estimates in an equilibrium framework which endogenizes candidate strategies. In this context, the estimation of supply parameters contributes new evidence to the literature on political common agency, where politicians must balance the competing demands of their party and constituents. Several papers have explored the impact of party discipline, special interest groups, and constituent preferences on politicians' positioning, with most concluding that

party discipline is both strong and increasing over time (Ansolabehere et al., 2001; Lee et al., 2004; Mian et al., 2010; Canen et al., 2020, 2026; Bombardini et al., 2023; Iaryczower et al., 2024; Cox and Shapiro, 2024). My framework contributes to this literature in two key ways. First, it directly measures candidates' sensitivity to voters' actual policy preferences, rather than relying on proxies like Presidential vote share. This approach allows for a direct comparison of how much weight candidates place on constituents' preferences versus party loyalty. Second, it analyzes candidate responsiveness to constituents and party discipline before the election, rather than focusing solely on post-election behaviors.

This article also contributes to the literature on measuring politicians' ideological positioning by developing a framework with four compelling features. First, it relies on information extracted directly from candidate campaign materials, rather than their actions once elected—such as roll-call votes (Poole and Rosenthal, 1985; Martin and Quinn, 2002), speeches in Congress (Gentzkow et al., 2019; Enke, 2020), or proxies like campaign contributions (Bonica, 2013; Hall and Snyder, 2015) or voters' behavior (Krasa and Polborn, 2014). Tausanovitch and Warshaw (2017) show these measures are often weakly correlated with each other and may not capture the actual ideological variation presented to voters. Second, by observing both candidates, rather than only elected members, I am able to capture the behavior of both opponents in each election—a critical component for estimating the strategic game of candidate competition. Previous studies using candidate survey data as this paper does (Ansolabehere et al., 2001; Shor and McCarty, 2011; Shor and Rogowski, 2018) were constrained by the limited sample size of survey respondents. By incorporating candidate websites alongside surveys, I overcome this limitation. Candidate websites have also been used in the literature to derive unidimensional measures of ideology (Di Tella et al., 2023; Meisels, 2023; Porter and Case, 2026). Third, this framework allows me to assign candidate positions on multiple political dimensions, which are interpretable and comparable over time due to repeated survey questions. Finally, by combining survey and website data, I estimate candidate ideology without relying on party labels, unlike most past work. Party-based measures often misplace extreme candidates, as they reflect a politician's centrality within their party rather than true ideological positioning (Noel, 2014, 2016). Since party dynamics and ideology co-evolve, they also prevent meaningful comparisons over time.

Leveraging panel data of precinct-level results, this paper provides new insights into the documentation of political realignment in the U.S. by utilizing official, granular election results. While previous studies (e.g., Kitschelt and Rehm (2019); Gethin et al. (2022); Kuziemko et al. (2026) among others) have relied on survey data to document the evolution of voting behaviors, I complement these *stated preferences* approaches by providing evidence from *revealed preferences* based on actual voting behaviors. Using precinct-level data also prevents biases in sample selection and allows me to bring more nuance to the extent of political realignment, along many dimensions and across various geographical scales.

The remainder of this paper is organized as follows: Section 2 outlines the conceptual framework of the paper, Section 3 presents the data, the methods to measure candidate ideology, and provides descriptive statistics, Section 4 describes the estimation of demand-side parameters, Section 5 describes the estimation

of the supply-side parameters, Section 6 decomposes changes in candidate positions and voting behavior between demand-side and supply-side factors, and Section 7 concludes and discusses the results.

2 Empirical Model

2.1 Setup

I consider a static model of local political competition where two candidates compete to maximize their probability of being elected while complying with party discipline.

The setup is characterized by two types of agents. In each district, indexed by d, for a given election year, indexed by t, there is a set of voters $\mathcal{I}_{dt}$, each indexed by i with observable characteristics $\mathbf{w_{idt}}$ (e.g., education, race, age). Voters choose between two candidates, D and R, each indexed by j, and the option of abstaining. Candidates choose their k-dimensional political platforms, $\mathbf{x_{jdt}} \in \mathbb{R}^k$, for $j = D, R$.

When voting for candidate j, voter i's utility is $u_{ijdt} = v_{it}(\mathbf{x_{jdt}}; \mathbf{w_{idt}}) + \varepsilon_{ijdt}$. When abstaining, voter i's utility is $u_{i0dt} = \varepsilon_{i0dt}$, where ε_{ijdt} are idiosyncratic i.i.d. mean-zero taste shocks. The probability that i chooses alternative j in district d in election t is then

$$s_{ijdt}\left(\mathbf{x_{Ddt}}, \mathbf{x_{Rdt}}; \mathbf{w_{idt}}\right) \quad = \quad \Pr\left(u_{ijdt} \geq \max_{l \in \{D,R,0\}} u_{ildt}\right). \tag{1}$$

The total vote share of each alternative j can be written as:

$$s_{jdt} = \int s_{ijdt}\left(\mathbf{x_{Ddt}}, \mathbf{x_{Rdt}}; \mathbf{w}\right) dF_{dt}(\mathbf{w}), \tag{2}$$

where $F_{dt}(\cdot)$ is the distribution of observable voter characteristics in district d in election t.

Throughout, lowercase vote shares denote model-implied objects, uppercase vote shares denote observed empirical counterparts, and tildes denote shares conditional on turnout.

The vector $\mathbf{x_{jdt}} \in \mathbb{R}^k$ is chosen by the local candidate ($j \in \{D, R\}$) trading off the probability of being elected on the one hand, and complying with the national leadership of their party on the other. Formally, candidates seek to be elected as they derive a rent $Q > 0$ from being in office, and maximize

$$\Pi_{jdt}(\mathbf{x_{jdt}}) = \underbrace{P_{jdt}(\mathbf{x_{jdt}}, \mathbf{x_{-jdt}})}_{\text{probability of winning}} Q - \underbrace{c_{jt}(\mathbf{x_{jdt}}, \mathbf{N_{jt}})}_{\substack{\text{cost of deviation from} \\ \text{national party platform}}} \tag{3}$$

where the probability of winning $P_{jdt}(\mathbf{x_{jdt}}, \mathbf{x_{-jdt}})$ depends on the vote share, conditional on turnout, $\widetilde{s}_{jdt}$ and a random shock, and where the function $c_{jt}(\cdot)$ captures the cost candidates incur when deviating from the national party position $\mathbf{N_{jt}} \in \mathbb{R}^k$. The lower the cost of deviating, the greater candidates' ability to adapt to their local conditions.

The need to comply with the position of the national leadership is not modeled explicitly, yet it is justified by a variety of political and institutional factors. First, party leaders reward candidates who follow the party line, both before and after the election. Second, it is costly for local candidates to come up with their own political strategy. I treat the evolution of the national leadership platforms as exogenous. Exogeneity here is intended as relative to the determinants of the local elections I study, and can be justified by the limited impact of each congressional district with respect to federal concerns.

A Nash equilibrium in district-election (d,t) is characterized by a collection of positions $(\mathbf{x_{Ddt}}, \mathbf{x_{Rdt}})$ and vote shares $(s_{Ddt}, s_{Rdt}, s_{0dt})$ such that (i) $\mathbf{x_{jdt}}$ maximizes $\Pi_{jdt}(\mathbf{x_{jdt}})$ and (ii) $s_{jdt} = \int s_{ijdt}\left(\mathbf{x_{Ddt}}, \mathbf{x_{Rdt}}; \mathbf{w}\right) dF_{dt}(\mathbf{w})$, for $j \in \{D, R, 0\}$.

2.2 Empirical Goal

The main objective of this paper is to decompose the change in voting behavior between two periods into changes in $s_{ijdt}(\cdot)$ and $F_{dt}(\mathbf{w})$ on the demand side and changes in $\mathbf{N_t} = (\mathbf{N_{Dt}}, \mathbf{N_{Rt}})$ and $c_{jt}(\cdot)$ on the supply side. Importantly, I evaluate the impact of changes in leadership positions exclusively through the resulting adjustments to $(\mathbf{x_{Ddt}}, \mathbf{x_{Rdt}})$ made by local candidates. This assumes away any direct influence that leadership changes might exert on local vote shares. Formally, the utility in (1) does not depend on $\mathbf{N_{jt}}$ directly.

To make these decompositions, I proceed in five steps. First, I employ granular precinct-level data to obtain precise measures of the observed vote shares S_{jdt}. Second, I obtain measures of $\mathbf{x_{jdt}}$ and $\mathbf{N_{jt}}$ using an ideal point model of candidate ideology. Third, I leverage variation in $\mathbf{x_{jdt}}$ between adjacent precincts in bordering districts to identify and estimate $s_{ijdt}(\cdot)$ conditional on t and voter characteristics $\mathbf{w_{idt}}$. Fourth, I recover the supply-side cost functions $c_{jt}(\cdot)$ underlying candidates' choices of positions $\mathbf{x}_{jdt}$, taking demand as given. Lastly, the parameters recovered in the first four steps allow me to conduct a number of counterfactual exercises. For example, I recover alternative candidate positions and voting behavior if the party leadership positions $\mathbf{N_t} = (\mathbf{N_{Dt}}, \mathbf{N_{Rt}})$ had not changed but all other factors changed, or if voter preferences $s_{ijdt}(\cdot)$ had not changed but all other factors had.

I first discuss the measurement of S_{jt}, $\mathcal{I}_t$, $\mathbf{x_t}$, and $\mathbf{N_t}$, and document the time series variation whose decomposition is my main object of interest. I then introduce an econometric specification for the estimation of the primitives $s_{ijdt}(\cdot)$ and $c_{jt}(\cdot)$, and discuss exclusion restrictions that are sufficient for identification.

3 Data and Descriptive Statistics

3.1 Panel of Precinct-level Election results

I collect precinct-level electoral results from 2000 to 2020. Precincts are the smallest geographical unit at which election results are available in the U.S. Each precinct has an average population of 1,000 registered

voters. There are about 175,000 precincts in the U.S., with on average 400 precincts per congressional district, which allows me to obtain considerable variation within congressional districts. Despite these election results being made public by county or state officials, there has so far been no unified dataset that contains the precinct-level Congressional results for the entire period that includes the geographical boundaries of each precinct. I therefore combined data from more than 100 different sources, such as Secretaries of State and county officials, in order to obtain results for the largest possible number of states over the period. The list of sources is described in the Appendix section dedicated to the description of the electoral data (see Appendix C).

Figure 1 shows the distribution of precincts in the U.S. for the 2020 election. Using precinct-level data instead of county-level data offers three advantages. First, precincts are much smaller and more demographically homogeneous than counties, reducing concerns about ecological fallacy and facilitating the estimation of individual preferences. Second, unlike counties, precincts are designed to have roughly equal populations everywhere, which allows for significant variation in both urban and rural areas. In contrast, using county-level voting data provides very scarce within-congressional district variation in urban areas. For instance, there are 16 congressional districts in Los Angeles county alone and Manhattan spans 3 distinct districts. In comparison, these counties have 4,312 and 1,266 precincts, respectively. The third advantage of precinct-level over county-level data is that the finer granularity offers a stronger foundation for identification assumptions, as described in Section 4.

Since precinct boundaries are changing over time, building a panel of electoral precincts is challenging. I therefore apportion all votes to the 2010 census block-groups, which have approximately the same population as electoral precincts. This offers the benefit of creating a consistent panel dataset with geographies that can be compared across elections. This is also the most granular level at which census demographic variables are available. Here and throughout, I use the term *precinct* instead of *block-group* as the unit of observation, with a minor abuse of terminology. Appendix C describes the strategy implemented to apportion the electoral results to the block-group level by using the spatial overlaps of precincts and census blocks (35 blocks per block-group). In order to maintain a balanced panel of states across the two periods of analysis, I include only those states that are available in both the pre-2010 and post-2010 periods. Table A.2 in the Appendix presents the sample of states included in the analysis.

I then combine election results with census demographics at the block-group level from NHGIS IPUMS (Manson et al., 2021) using the decennial census and the American Community Survey (ACS). In order to estimate individual-level preferences and not only aggregate preferences, I recover the full distribution of demographics along three dimensions: education, race, and age. This requires having both the average, the variance, and the covariance of each of these demographic variables at the block-group level. Since only marginal distributions are reported in the decennial census and the ACS, I recover the joint distribution at the block-group level from a multi-scale model combining block-group level and tract-level demographic counts with PUMA-level individual data from the ACS individual files. Appendix D details the strategy

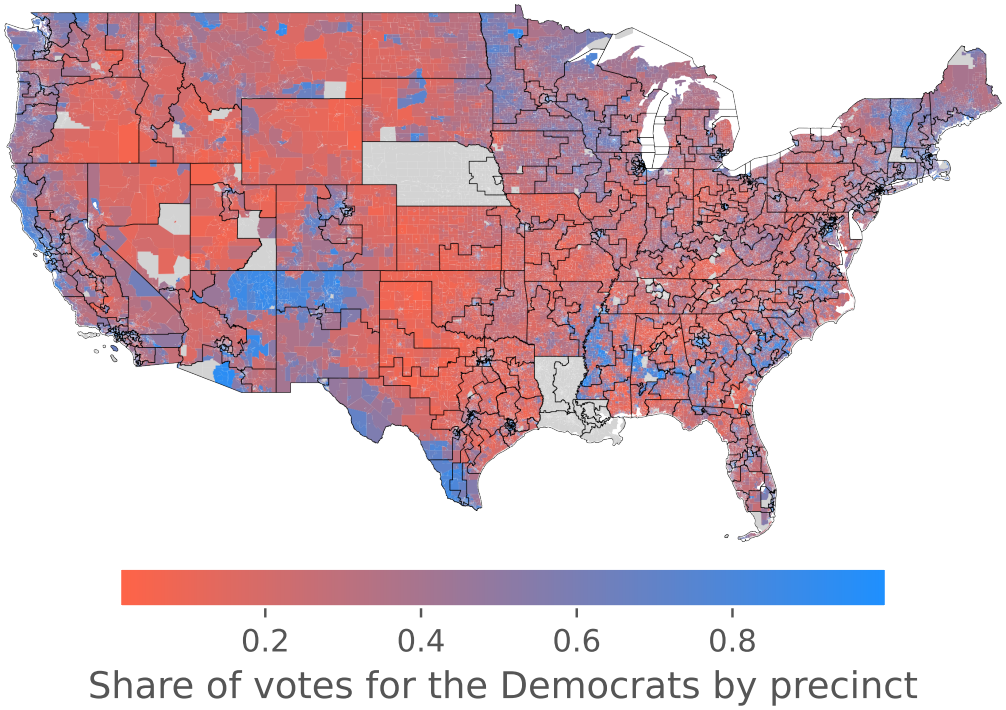
to recover the full distribution of demographics in each block-group.

In the remainder of this paper, I refer to white and non-white voters with a specific definition. The census asks both about race and ethnicity, and people can identify as belonging to multiple races. In order to have demographic groups with sufficiently large sizes in most precincts, I classify both "Hispanic white" and "Hispanic non-white" as "non-white" and individuals reporting multiple races as "non-white". The group of "white" voters therefore consists of "Non-Hispanic whites," with a single race. Under this definition, the average share of whites is 75% in 2000 and 68% in 2020.

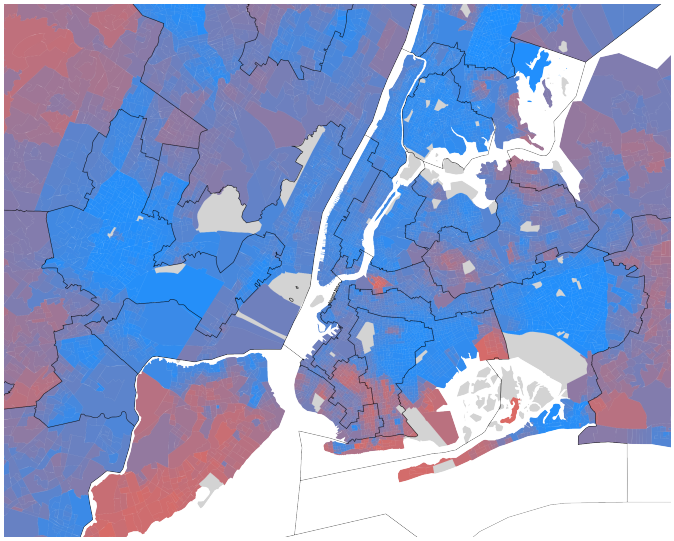
For each precinct, I define turnout as the ratio of total votes cast in the election to the population aged 18 or older.[6]

In addition to census demographics, I also collect the share of unionized workers using localized contract-level data from the Federal Mediation and Conciliation Service (FMCS), which gives establishment-level data on unionization. I geocode each establishment to obtain its census block. I then apportion employment from the workplace to the voting place using LEHD Origin-Destination Employment Statistics (LODES) data, which give yearly block-to-block job flows. This strategy allows me to obtain a block-group-level estimate of voters who are members of a labor union. I also collect the share of voters that belong to any church, and specifically that belong to an evangelical church, using a combination of data from the Census of Religious Bodies and data from Axle.

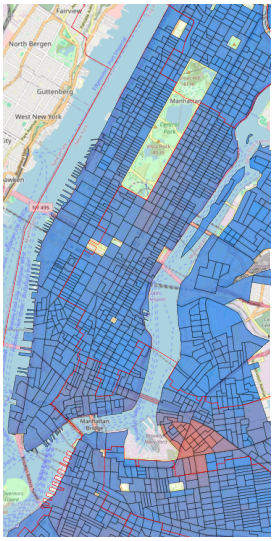
[6]The turnout denominator could be refined by excluding non-citizens, but census estimates of citizenship status are noisy and would add measurement error to a key variable. Because all specifications include precinct fixed effects, any resulting bias should be absorbed and is therefore unlikely to affect the results.



(a) Continental United States



(b) New York City, NY



(c) Manhattan, NY

Figure 1: Precinct-level Democratic vote share in 2020

Notes: The figure shows the 2020 distribution of House Democratic vote share in each precinct, conditional on turnout. Congressional district borders are shown in black (panels (a) and (b)) and in red (panel (c)). Each congressional district contains on average 400 precincts. Precinct results have been spatially interpolated at the block-group level as described in Appendix C.

3.2 Evidence of Political Realignment

I start by providing new evidence of political realignment using this novel panel of precinct-level electoral results. Most of the existing evidence on realignment has been documented through survey data (Kitschelt and Rehm, 2019; Gethin et al., 2022; Kuziemko et al., 2026). Using aggregate election results presents three main advantages. First, it allows me to study the revealed rather than stated preferences of voters, which may more accurately reflect their voting choices, as survey respondents might misreport their votes. Second, it helps reduce potential sample bias resulting from the partial coverage of surveys, which may not reach the entire voter population. Lastly, it gives more power to study precise phenomena and detect non-linearities. Naturally, relying on aggregate results also has some drawbacks. The main issue is to recover individual voting behavior from aggregate election results, referred to as the ecological inference problem (King, 1997). I discuss these challenges and the strategy proposed to overcome them in Section 4.

While much of the literature on political realignment has focused on education, I begin by examining the variables along which this realignment has been most pronounced. Specifically, I analyze twelve demographic variables capturing precinct-level demographic composition, occupational distribution, and affiliation with religious or labor unions. To compare the extent of realignment across these dimensions, I run a single linear regression of precinct-level Democratic vote shares on demographic variables interacted with a time trend. Each demographic variable is normalized to have a mean of zero and a standard deviation of one to ensure comparability of coefficients across dimensions. The regression specification is as follows:

$$\widetilde{S}_{Dpt} = \sum_{m=1}^{M} \beta_m w_{m,pt} \cdot t \; + \; \sum_{m=1}^{M} \gamma_m \, w_{m,pt} \; + \; \mu_t + \varepsilon_{pt} \tag{4}$$

where $\widetilde{S}_{Dpt}$ is the House Democratic vote share in precinct p at election t, conditional on turnout, each $w_{m,pt}$ captures demographic dimension m of precinct p in year t, μ_t are election fixed effects. Figure 2 reports the estimates from the joint regression including all demographic variables, Appendix Figure A.3 reports the unconditional estimates from running a separate regression for each demographic variable. Positive values ($\beta_m > 0$) indicate a realignment toward the Democratic Party while negative values indicate a realignment toward the Republican Party. The figure shows that education dominates all other demographic variables in absolute value, with more-educated areas shifting toward the Democrats. Realignment along education lines is three times as large as realignment along income lines. Other variables also matter, especially the share of white residents and the share of residents living in rural areas, with predominantly white and more rural areas shifting toward the Republicans. Appendix Figure A.3 shows that similar results emerge from separate regressions for each demographic variable. Appendix Figure A.4 shows that the same pattern holds when demographic composition is fixed at its 2000 level and when the analysis uses only within-precinct changes in demographics. In sum, education is the strongest

demographic correlate of realignment, both conditionally and unconditionally, and both across precincts that were initially more educated in 2000 and across precincts that became more educated between 2000 and 2020.

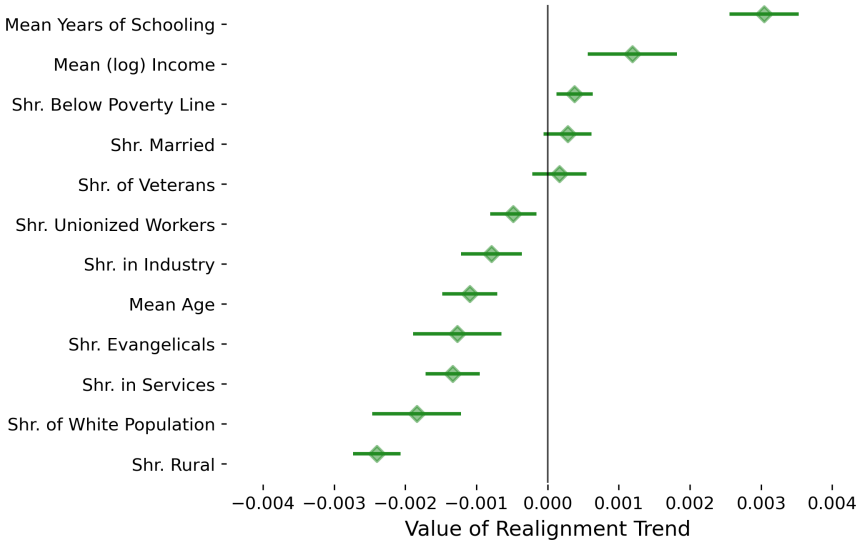


Figure 2: Observed Trends in Political Realignment: Education as the Key Driver

Notes: The figure shows, for each demographic dimension m, the coefficients β^m from the following linear regression: $\widetilde{S}_{Dpt} = \sum_{m=1}^{M} \beta_m \, w_{m,pt} \, t \;+\; \sum_{m=1}^{M} \gamma_m \, w_{m,pt} \;+\; \mu_t + \varepsilon_{pt}$ where $\widetilde{S}_{Dpt}$ is the House Democratic vote share in precinct p at time t, conditional on turnout, $w_{m,pt}$ captures demographic dimension m, and μ_t are election fixed-effects. Positive coefficients indicate realignment toward the Democratic Party while negative coefficients indicate realignment toward the Republican Party. The bars around each marker show the 95% confidence intervals with standard errors clustered at the congressional-district-by-year level. Appendix Figure A.3 shows the results from a regression of each demographic variable separately. Appendix Figure A.4 shows the results from regressions holding demographics at their 2000 level and controlling for precinct fixed effects.

I now turn to measures of political realignment focusing on educational lines. Figure 3 shows, for each election, the linear relationship between precinct-level education and Democratic voting. In each successive election, the average gap in education between Democratic and Republican voters has increased by 2.5 months of schooling. Importantly, the rise in this correlation preceded the election of Donald Trump in 2016, with no specific jump in the educational gradient in that year.

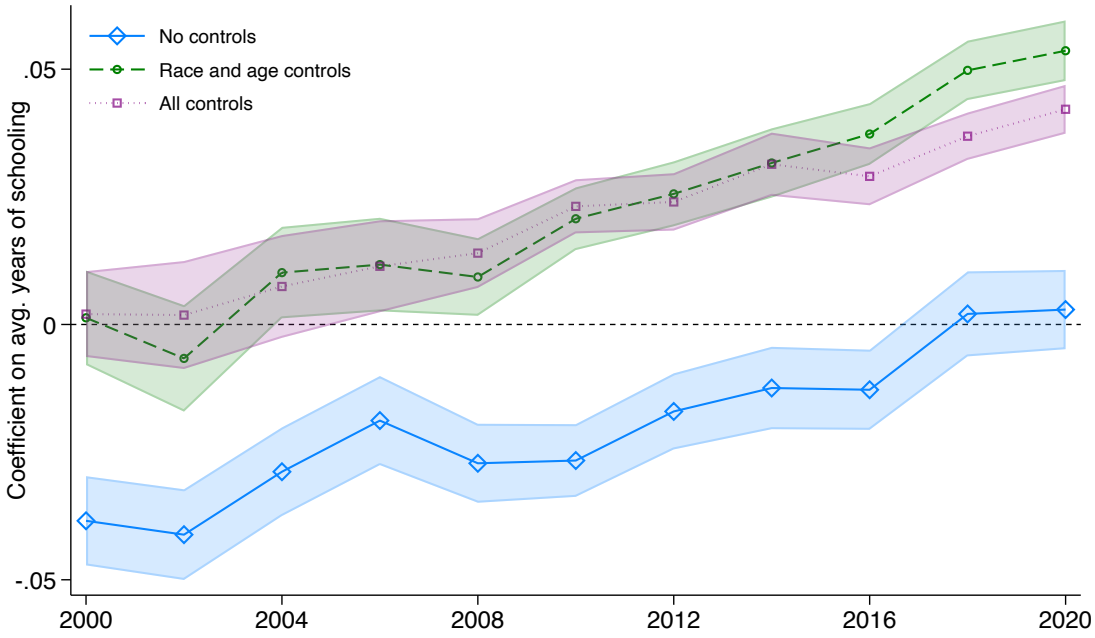


Figure 3: Increasing correlation between precinct-level education and Democratic vote shares.

Notes: The figure shows the coefficients β_t^{edu} from the following regression: $\widetilde{S}_{Dpt} = \beta_t^{edu} edu_{pt} + \mu_t + \epsilon_{pt}$. The coefficients in blue with diamond markers are unconditional while the coefficients in green with circle markers are from a regression controlling for the precinct-level share of white and the precinct-level average age by year. The coefficients in purple with square markers additionally control for all the demographic controls, as in Figure 2, separately for each year. Standard errors are clustered at the congressional-district-by-year level

The fine-grained nature of precinct-level election results allows me to examine this relationship in more detail and investigate potential non-linearities. Figure A.5 shows the average Democratic vote share for each percentile of the education distribution. The figure shows a U-shaped relationship between education and Democratic vote shares for both the early and later years, with most changes over time occurring at the tails of the distribution. Figure A.6 also shows the evolution of the education gradient within congressional districts. The relationship within districts is strongly negative, with more-educated precincts inside each congressional district voting more for the Republican candidates, with little change over time, echoing findings from Gelman (2009) for income. Overall, the precinct-level U-shaped relationship can be thought of as the sum of two opposite relationships, between and within districts.[7]

3.3 Candidate ideological positioning

In order to recover candidate ideologies on multiple dimensions, I combine candidate survey data from Project VoteSmart with text data from candidate websites obtained from the United States Elections

[7]For each year, the total covariance between education and Democratic vote share $\mathbb{C}(\widetilde{S}_{Dpt}, edu_{pt})$ can be decomposed into the within district covariance $\mathbb{E}\big[\mathbb{C}(\widetilde{S}_{Dpt}, edu_{pt}|d(p))\big]$ and the between district covariance $\mathbb{C}\big[\mathbb{E}(\widetilde{S}_{Dpt}|d(p)), \mathbb{E}(edu_{pt}|d(p))\big]$. For instance, in 2020, $\mathbb{C}(\widetilde{S}_{Dpt}, edu_{pt}) = 0.005$, $\mathbb{E}\big[\mathbb{C}(\widetilde{S}_{Dpt}, edu_{pt}|d(p))\big] = -0.012$, and $\mathbb{C}\big[\mathbb{E}(\widetilde{S}_{Dpt}|d(p)), \mathbb{E}(edu_{pt}|d(p))\big] = 0.017$.

Web Archive, maintained by the Library of Congress.[8] Appendix Figures A.1 and A.2 provide illustrative examples of the raw data.

Project VoteSmart is a nonpartisan organization that has surveyed candidates since the 1990s, covering a wide range of questions on candidates' policy positions across various topics. Response rates began to decline in the late 2000s, prompting Project VoteSmart to conduct internal research using candidate statements, press releases, and interest group ratings to impute answers for candidates who have not responded. When candidates' positions on an issue remained unclear from these sources, the response was left unknown.

Because some questions vary across election cycles, I match questions across cycles only when their wording is nearly identical. I include questions asked in at least two election cycles and exclude those focused on specific politicians or on country-specific foreign-policy issues. From 2000 to 2020, this leaves 132 distinct questions, which fall into 15 topics: abortion, crime, education, environment, gun regulation, campaign finance, immigration, international relations and security, diversity, employment, trade, taxes, health care, social security, and welfare. I classify these topics into two broad dimensions, which I interpret as cultural and economic issues. Employment, trade, taxes, health care, social security, and welfare are classified as economic issues; all other topics are classified as cultural issues.

To assess the validity of this classification, I re-estimate candidates' positions separately for each topic and compute their correlations with the two broad dimensions, excluding that topic from the construction of the aggregate dimensions. Appendix Figure A.8 reports these correlations. Most topics load clearly on one of the two dimensions: topic-specific responses are much more strongly correlated with one aggregate dimension than with the other. Environmental issues are the most balanced across dimensions, with some heterogeneity across parties: Democratic environmental positions are more strongly correlated with cultural positions, while Republican environmental positions are more strongly correlated with economic positions. Because the average loading is stronger on the cultural dimension, I classify environmental issues as cultural.

I estimate each candidate's time-varying ideal point on every ideological dimension using a dynamic Bayesian item-response model, similar to Clinton et al. (2004), Jessee (2009), and Shor and Rogowski (2018), detailed in Appendix Section E. For each candidate j in district d at election t, I observe responses to Q survey questions. Most items are binary, while a small subset has ordered response categories. I allow each candidate to have a time-varying ideal point x_{jdkt} on dimension k, evolving as $x_{jdkt} = \rho_k x_{jdk,t-1} + \sigma_k \eta_{jdkt}$. The scale is normalized so the prior for x_{jdkt} is stationary with unit variance. For binary items, I use a standard two-parameter logistic model with an item cut-point, β_q and an item discrimination parameter, α_q. The cut-point captures the ideological position at which the "yes" and "no" responses are equally likely; the discrimination parameter captures how sharply the probability of a "yes" changes as a candidate's ideal point moves away from the cut-point. For ordered items, I use

[8] Websites are only available starting in 2002.

a graded-response model with an item discrimination parameter and a set of ordered cut-points that partition the latent ideological scale into response categories. All model parameters are estimated with Bayesian Markov chain Monte Carlo implemented in PyMC. Responses that are not observed are excluded from the likelihood. For each candidate-period ideal point, I take the posterior mean as the estimate and use the posterior standard deviation of the MCMC draws as its standard error.

To complement these survey-based ideal points for candidates who did not respond or provided only partial survey answers, I also incorporate data from candidate websites. Using the United States Elections Web Archive, I scrape each candidate's website as archived on the day before the general election. I then process the text and transform it into embedding vectors that provide numerical representations of candidates' campaign content (Dai et al., 2015), as detailed in Appendix Section E. I then train a machine-learning regressor of survey-based ideal points on the text features, using Extreme Gradient Boosting (Chen et al., 2015). The model achieves a mean squared error of 0.17 for the economic dimension and 0.21 for the cultural dimension.

Finally, I combine the information from surveys and websites using the relative uncertainty of both measures. Candidates without any survey responses are assigned their website ideal points only ($x_{jdkt}^{\text{website}}$), similarly, candidates for whom I do not observe the website are assigned their survey ideal point only. For all those with both survey and website ideal points, I take a weighted average of the two measures using their relative uncertainty as the weighting factor:

$$x_{jdkt} = \omega_{jdkt} x_{jdkt}^{\text{survey}} + (1 - \omega_{jdkt}) x_{jdkt}^{\text{website}}, \tag{5}$$

with $\omega_{jdkt} = \frac{MSE_k^{website}}{se(x_{jdkt}^{\text{survey}})^2 + MSE_k^{website}}$.

Appendix Figure A.16 shows the correlation between the survey-based x_{jdkt}^{survey} and website-based ideal points ($x_{jdkt}^{\text{website}}$) on each dimension, which is above 90% overall and above 70% within party. I do not observe the ideology of about 12% of the sample of candidates, who are therefore excluded from the analysis.

Figure 4 shows the distribution of candidate positions on the cultural and economic dimensions. The figure shows the positions of notable candidates in 2020, along with the positions of party leadership, which are averages of the ideological positions of key party figures: the Speaker of the House, the Majority and Minority Leaders, the Whips, and the Chairs of the Caucus and the Conference.

The distributions of Democratic and Republican candidates are clearly distinct from each other. However, contrary to measures such as DW-Nominate, the distributions of candidates from the two parties overlap substantively, with the most conservative Democratic candidates located to the right of the most progressive Republican candidates. Canen et al. (2026) found similar polarization patterns in Congress, once accounting for party discipline from party leaders.

Importantly, while the overall Spearman (rank) correlation between the cultural and economic dimen-

sions is 0.89, the within-party correlations are much smaller: 0.57 among Democratic candidates and 0.41 among Republican candidates. These relatively low correlations suggest that candidates' cultural and economic positions vary independently, with some candidates substantially more progressive on one dimension than on the other.

Appendix Figure A.7 compares the ideology measures developed in this paper with commonly used measures of ideology, such as DW-Nominate (Poole and Rosenthal, 1985), two-dimensional positions from Canen et al. (2026), which are both available for election winners only, and the contribution-based measure of ideology from Bonica (2014), which is also available for election losers. Both the cultural and economic dimensions are highly correlated with these existing measures, and the correlations are similar in magnitude to those observed among the existing measures themselves.

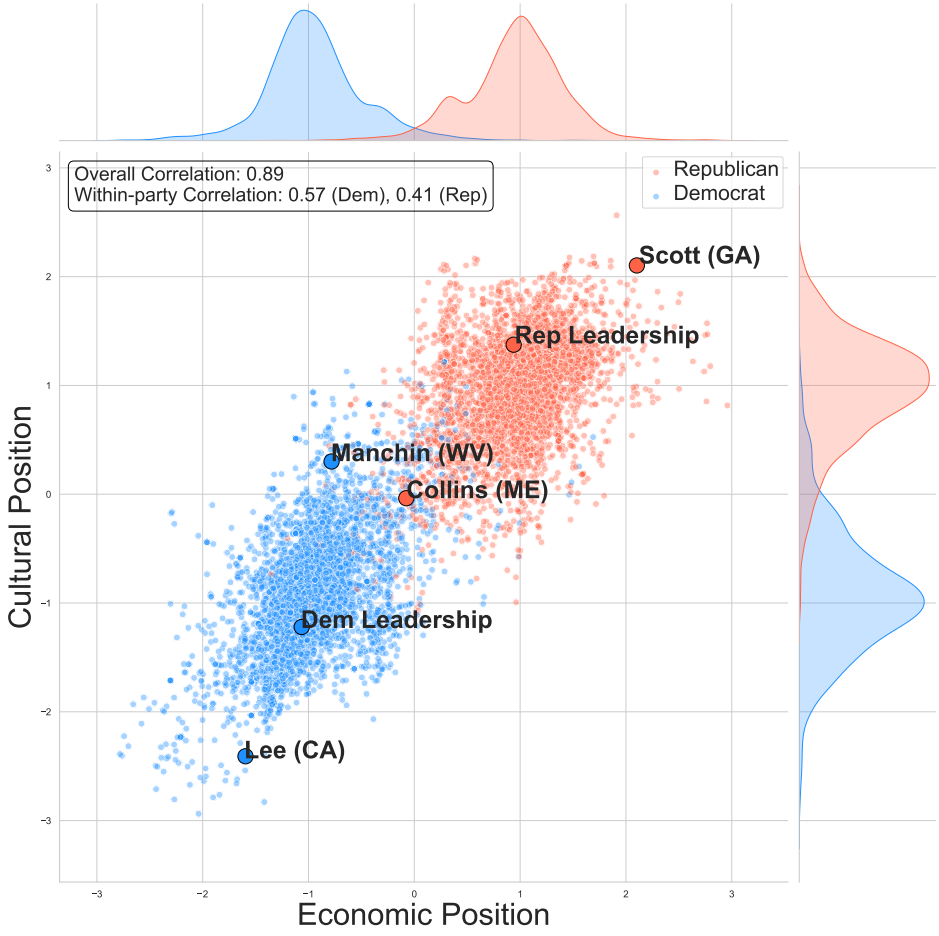


Figure 4: Distribution of candidate ideal points

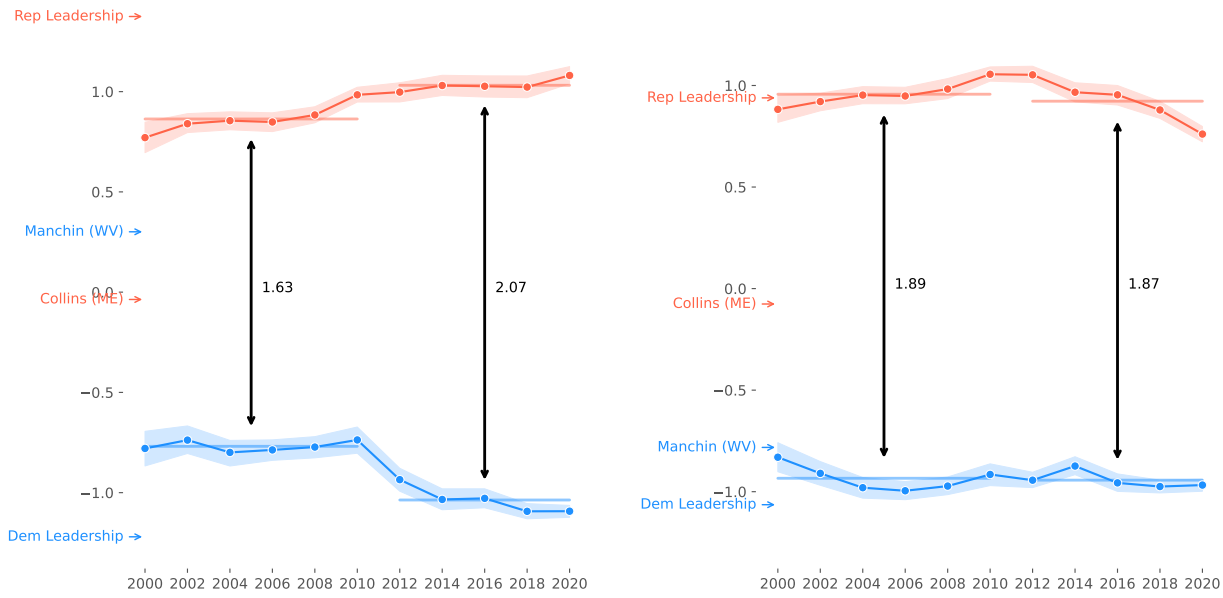
Notes: Each dot shows one congressional candidate's estimated two-dimensional ideal point from the multimodal text-and-survey model on cultural and economic issues for each election. Notable candidates' positions as well as the party leadership position in 2020 are added to the graph. The marginal densities are plotted on each axis. The graph excludes third parties and independent candidates. The correlation between the two dimensions is 0.89 in the whole sample and the within-party correlations are 0.57 for the Democratic Party and 0.41 for the Republican Party.

3.4 Evidence of Polarization

Figure 5 shows the evolution of candidate ideal points on both cultural and economic issues over time. The average distance between the two parties has dramatically increased on cultural issues, while it has stayed somewhat constant on economic issues. The average distance between a Democratic and a Republican candidate on cultural issues has increased by about half of a standard deviation in ideology between 2000 and 2020.

The overall dynamics of polarization align with similar findings from other types of data (roll-call votes, speeches in Congress, campaign contributions, etc.). To the best of my knowledge, this paper is the first to document a clear contrast in polarization, with cultural issues showing significantly more divergence than economic issues. Appendix Figure A.10 shows the same relationship separately for winning and losing candidates.

Notably, the rise in political polarization between the two main parties has almost mechanically led to an increase in the overall correlation between candidate cultural and economic ideal points. However, when looking within parties, the correlation between candidate positions across the two dimensions has decreased over time. In 2000-2010, candidates' cultural and economic positions had a 0.62 Spearman correlation coefficient for Democratic candidates and a 0.48 coefficient for Republican candidates. This rank correlation declined to 0.43 for Democratic candidates and 0.28 for Republican candidates in 2012-2020. This decline in the correlation indicates that as the two parties moved further apart, candidates began to differentiate themselves more across the two dimensions.



(a) Cultural Positions

(b) Economic Positions

Figure 5: Larger Polarization on Cultural vs. Economic Issues

Notes: The figure shows the evolution of average candidate positions by party and dimension, with 95% confidence intervals around the mean. Appendix Figure A.10 shows the same relationship separately for winning and losing candidates. The figure also reports the average distance between Democratic and Republican candidates in each period. Party leadership positions and notable candidates' positions in 2020 are added to both panels.

3.5 Determinants of candidate positioning

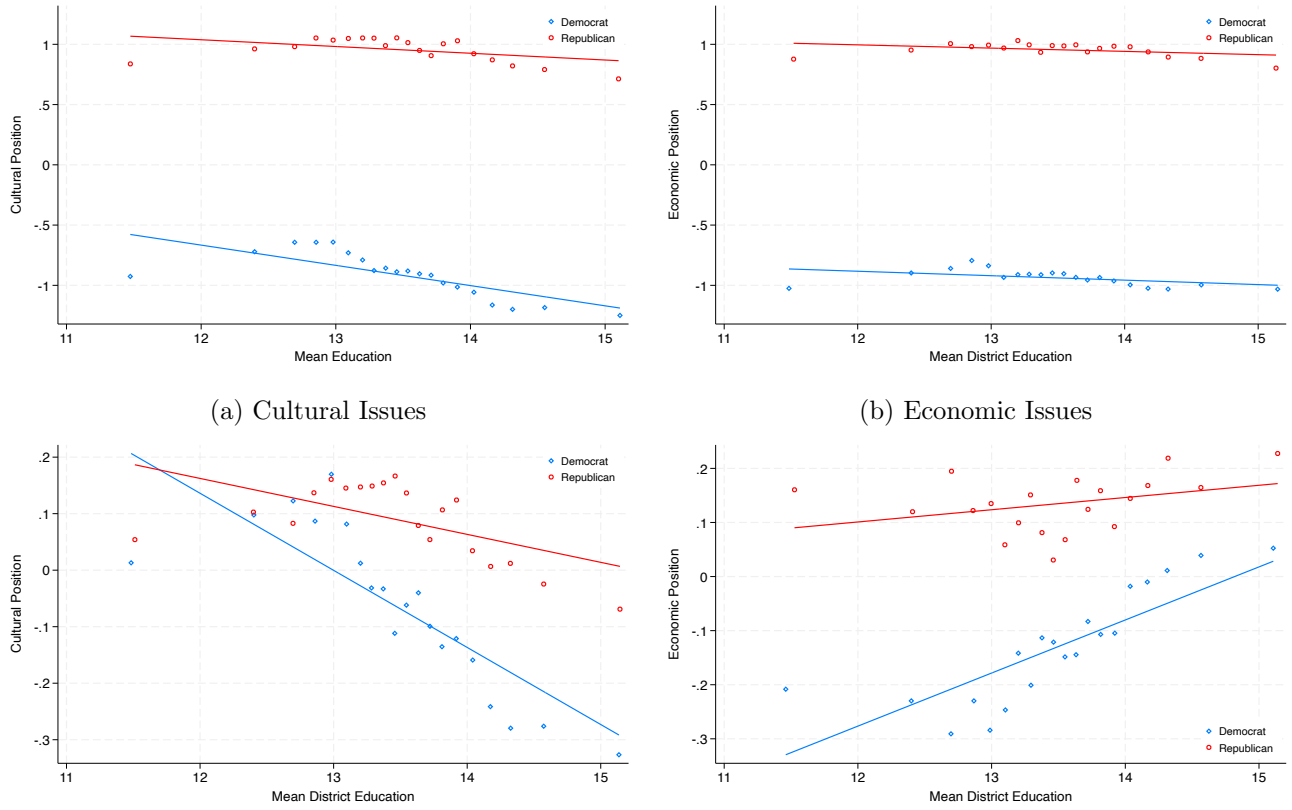
This section examines the extent to which the demographic composition of congressional districts correlates with candidates' ideal points. Based on classic spatial models, such as Downs (1957), and the Median Voter Theorem (Black, 1958), one would expect candidates to adapt their positions to align with their constituents' ideology, resulting in significant ideological variation across districts.

Figure 6 illustrates these ideological differences by focusing on the educational composition of districts. For each 5-percentile bin of the education distribution, it shows the average positions of Democratic and Republican candidates. The top two panels, which display unconditional relationships, reveal substantial differences in candidate positions based on district education levels. On cultural issues, Democratic candidates in the most educated districts tend to be much more progressive than those in the least educated districts. Republican candidates exhibit a similar, though slightly less pronounced, pattern. By contrast, panel (b) shows that there is almost no difference in candidate positions on economic issues across education levels in these unconditional graphs.

The bottom two panels display the same relationships, but controlling for candidate position on the other dimension. The negative relationship between candidate cultural positions and district education holds true for both Democratic and Republican candidates. The economic dimension exhibits the opposite

pattern: both Democratic and Republican candidates tend to offer more conservative economic positions in more-educated districts, once accounting for their cultural positions. Appendix Figure A.9 presents similar results for additional demographic variables.

All these patterns provide suggestive evidence that candidates might be adjusting their positions to their constituents' preferences. Section 5 precisely estimates the weight that candidates allocate to their constituents in their choice of positions.



(a) Cultural Issues

(b) Economic Issues

(c) Cultural Issues, controlling for Economic

(d) Economic Issues, controlling for Cultural

Figure 6: Candidate positions and congressional district composition

Notes: Each panel shows the relationship between candidate positions and the district average level of education. Each dot represents 5% of the education distribution and shows the average position of candidates, separately for Democrats and Republicans, for the districts in this quantile. Panels (a) and (b) present unconditional candidate positions. Panels (c) and (d) show the same relationship but controlling for the position on the other dimension. For example, on panel (a), Democratic candidates running in the most educated districts tend to be 0.5 points more progressive (1 standard deviation) than Democratic candidates running in the least educated districts. Figure A.9 shows the same relationship for other demographic variables.

4 Estimation of voter preferences

4.1 Model

The model of vote choice features individual voters with heterogeneous preferences over candidate characteristics. Voters make a choice between a Democratic candidate, a Republican candidate, and abstaining, which I model as resulting from maximizing a random indirect utility with heterogeneous preferences. I give the following functional form to the utility of voter i, who resides in precinct p, itself located in district d, when choosing candidate $j \in \{D, R\}$ in election t:

$$u_{ijpt} = \sum_k \beta_{ikt} x_{jdkt} + \alpha_{ijt} + \xi_{jpt} + \varepsilon_{ijpt}, \tag{6}$$

where x_{jdkt} is the position of candidate j in district d on dimension k in election t, α_{ijt}, $j \in \{D, R\}$ is voter i's utility of voting for candidate j's party, independently of their positions, ξ_{jpt} is a party-specific precinct-level taste shock, and ε_{ijpt} is an idiosyncratic voter-candidate taste shock, which I assume follows a type-I extreme value distribution. The mean utility of abstaining is normalized to zero, so $u_{i0pt} = \varepsilon_{i0pt}$.

To capture the heterogeneity in voter preferences over ideology and over partisan preferences, I decompose voter preference parameters as:

$$\alpha_{ijt} = \alpha_{jt} + \mathbf{w}'_{it} \boldsymbol{\alpha}^{\mathbf{w}}_{jt}, \tag{7}$$

$$\beta_{ikt} = \beta_{kt} + \mathbf{w}'_{it} \boldsymbol{\beta}^{\mathbf{w}}_{kt} + \sigma^{\beta}_{kt} \nu_{kit}, \tag{8}$$

for each ideological dimension k and for each party j. $\mathbf{w}_{it}$ is a vector of observed voter characteristics (years of schooling, race, and age), $\boldsymbol{\alpha}^{\mathbf{w}}_{jt}$ and $\boldsymbol{\beta}^{\mathbf{w}}_{kt}$ are vectors of dimension $|\mathbf{w}|$ capturing how voter partisan and ideological preferences vary with demographics. The term ν_{kit} captures residual individual heterogeneity in ideological preferences, with $\boldsymbol{\nu}_{it} \sim P\nu = \mathcal{N}(0, I_K)$ and scale parameter σ^{β}_{kt}. The distribution of demographics in each precinct p and election t is denoted by $F_{pt}(\cdot)$.

The share of votes received by candidate j in each precinct p is obtained by integrating over both demographic characteristics and unobserved heterogeneity:

$$s_{jpt} = \int \int \frac{\exp\left(\sum_k \beta_{ikt} x_{jdkt} + \alpha_{ijt} + \xi_{jpt}\right)}{1 + \sum_{l \in \{D,R\}} \exp\left(\sum_k \beta_{ikt} x_{ldkt} + \alpha_{ilt} + \xi_{lpt}\right)} dP_\nu \, dF_{pt}(\mathbf{w}). \tag{9}$$

Following Berry et al. (1995), I estimate the model with a GMM procedure that matches the model-

implied precinct vote shares to the observed shares.[9][10]

4.2 Identification

I first discuss the identification of voter preferences over candidates' endogenous characteristics (ideology) using a subset of electoral precincts located near congressional district borders. I then discuss how I recover the remaining demand-side parameters of the model using the full sample.

First Step The first step recovers the parameters $\boldsymbol{\beta}$ which capture heterogeneous voter preferences over candidate positions. Since candidate positions are likely correlated with voters' taste shocks, directly comparing vote shares across districts with different candidate positions would lead to biased demand estimates. To address this endogeneity concern, I propose an identification strategy that controls for unobservables affecting voters by exploiting the fine-grained structure of election data and the hierarchical nature of political competition. Candidate positions are chosen at the congressional-district level, while vote shares are observed at the precinct level. Since each district contains about 400 precincts, any individual precinct accounts for only a small share of the electorate relevant to candidates' positioning decisions. These decisions should therefore mainly reflect district-level electoral conditions, making it unlikely that candidates tailor their positions to precinct-specific taste shocks. The identification strategy exploits differences in candidate positions across congressional district borders, comparing adjacent precincts that face different candidates but should otherwise have similar voter characteristics and local taste shocks.

To recover voters' preferences over candidate positions, I decompose the precinct-level unobserved taste shock into three components: $\xi_{jpt} = \xi_{jg(p)t} + \xi_{jp} + \widetilde{\xi}_{jpt}$.

The first component, $\xi_{jg(p)t}$, is a party-specific taste shock that is common, at election t, to precincts in the same border group $g(p)$ on both sides of the district border. The second component, ξ_{jp}, captures time-invariant unobservables specific to each precinct-party pair.[11] The third component, $\widetilde{\xi}_{jpt}$, is the remaining precinct-party-election residual shock. The identifying assumption is given by:

$$\mathbb{E}[x_{jdkt} \cdot \widetilde{\xi}_{jpt}] \quad = \quad 0 \tag{10}$$

This assumption follows the reasoning by which differences in candidate positions across neighboring districts are not correlated with temporary, unobserved differences between adjacent precincts. This assumption is supported by the fact that each precinct has only a negligible impact on candidate vote

[9] I implement this estimation using the Python package provided by Conlon and Gortmaker (2023).

[10] As an alternative specification, reported in the appendix, I impose voter homogeneity within precincts, $\mathbf{w}_{it} = \mathbf{w}_{pt}$, and estimate a more parsimonious multinomial logit model without random coefficients, following Berry (1994). The two specifications yield similar results.

[11] I estimate two separate models, one for 2000–2010 and one for 2012–2020, each with its own set of precinct fixed effects.

shares, with an average of 400 precincts per district, and by the absence of systematic differences at district borders in non-electoral characteristics. Additionally, I only use district borders within the same state, ensuring no systematic differences in legislation or upper-level offices on either side of the district boundary. Congressional districts are obviously not drawn at random and state legislators might be aware of some precinct-level unobserved taste shocks when deciding which precincts to include in each district. For the identifying assumption to hold, *temporary* deviations between precincts on both sides of the border should not lead to changes in candidate positions. Appendix Figure A.11 shows that while in an OLS specification, precinct demographics are strongly correlated with candidate positions, this correlation disappears after conditioning on precinct-group-by-party-by-election and precinct-by-party fixed effects. The figure reports coefficients from specifications relating candidate positions to precinct-level census demographics, using only the sample of border precincts. One out of 20 coefficients is significant at the 10% level, as would be expected.

Figure 7 illustrates the precinct units located along congressional district borders and the border groups used in the analysis. For each state-year, I consider each pair of congressional districts that shares a common boundary. Within each district pair, I retain all precincts that intersect with that shared boundary. Because each such precinct can be adjacent to several precincts on the opposite side of the border, I do not construct simple one-to-one pairs. Instead, I construct contiguous cross-border groups of maximum size 5.[12] To determine these groups, I use an assignment algorithm that, within each district pair, selects geographically contiguous cross-border groups that minimize the distance between precincts assigned to the same group. Appendix C.4 provides more detail on the construction of precinct-groups.

The median number of precincts in each of these precinct-groups is 3 and 16% of the country's precincts are adjacent to a congressional district border. This strategy, which leverages spatial discontinuities, has been used in past studies to estimate the impacts of minimum wage (Card and Krueger, 1994; Dube et al., 2010), school valuations (Black, 1999), and advertising effects (Spenkuch and Toniatti, 2018).

[12] Dube et al. (2010) and Spenkuch and Toniatti (2018) create datasets at the county-pair level, where each county appears once for each contiguous neighbor. When treatment effects are assumed homogeneous and standard errors are clustered at the state-pair level (or district-pair level in my case), duplicating observations does not affect the estimates or their standard errors. In my setting, however, treatment effects are highly heterogeneous and precincts are much smaller than counties. It is therefore more reasonable to keep the number of observations fixed and construct groups rather than all possible pairs.

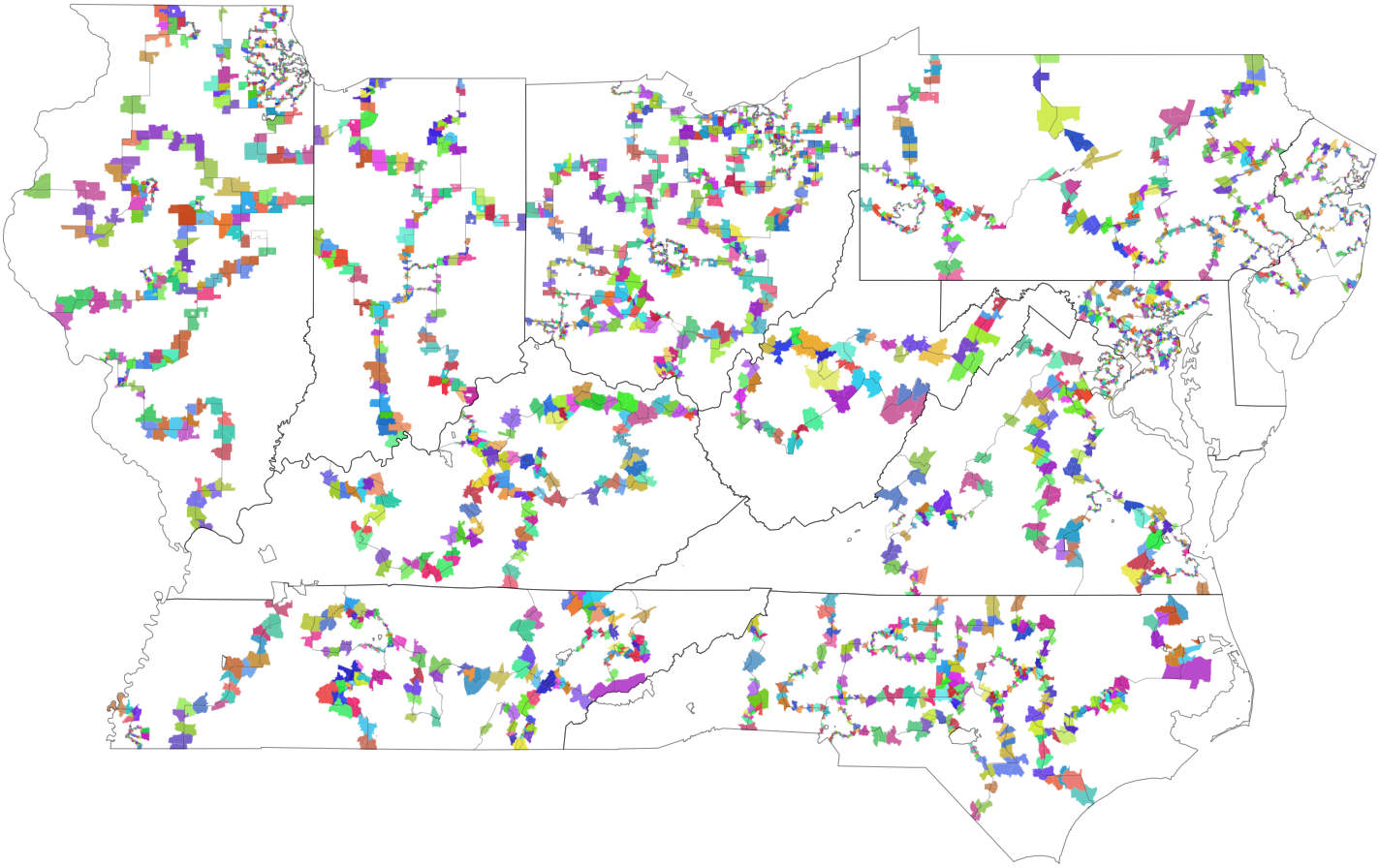


Figure 7: Precinct-group distribution for the 2020 elections (117th Congress).

Notes: The figure shows the precincts (block-groups) that sit on the district border. I use only borders within the same state and therefore exclude at-large districts (e.g., Delaware on the map). Appendix Section C details the construction of the precinct groups.

The moments used for the identification of the parameters in the first step, for each political dimension $k \in \{cultural, economic\}$, for each demographic characteristic w_{pt} (years of schooling, race, age, and interactions), for each party $j \in \{D, R\}$ are:

$$
\begin{aligned}
\mathbb{E}[x_{jdkt} \cdot \widetilde{\xi}_{jpt}] &= 0 \\
\mathbb{E}[\left(x_{jdkt} \cdot w_{pt}\right) \cdot \widetilde{\xi}_{jpt}] &= 0 \\
\mathbb{E}[w_{pt} \cdot \widetilde{\xi}_{jpt}] &= 0 \\
\mathbb{E}[x_{jdkt}^2 \cdot \widetilde{\xi}_{jpt}] &= 0 \\
\mathbb{E}[\left(x_{jdkt}^2 \cdot w_{pt}\right) \cdot \widetilde{\xi}_{jpt}] &= 0 \\
\mathbb{E}[\left(x_{Ddkt} - x_{Rdkt}\right) \cdot \widetilde{\xi}_{jpt}] &= 0 \\
\mathbb{E}[\left((x_{Ddkt} - x_{Rdkt}) \cdot w_{pt}\right) \cdot \widetilde{\xi}_{jpt}] &= 0
\end{aligned}
\tag{11}
$$

Table A.1 in the Appendix presents Monte Carlo simulations for the demand specification, providing evidence that the key coefficients are identified in this setting. The table compares the random-coefficients multinomial logit estimated at the precinct level with a more parsimonious standard logit specification with homogeneous voters, following Berry (1994), as well as the corresponding specifications estimated at the county level. The simulations show that both omitting individual-level preference heterogeneity and relying on more aggregated county-level data prevent the model from properly recovering the demand coefficients.

Second Step By design, the first step removes any identifying variation for characteristics that lack a discontinuity at the district border, such as precinct demographics ($w_{pt} \approx w_{g(p)t}$). This setup leaves voters' party preferences (α_{ijt}) largely unidentified in the first step.[13]

Given that a candidate's party affiliation is considered exogenous, it is not necessary to limit the analysis to precincts at the district border to identify party preferences. Consequently, in the second step, I utilize the entire sample, not just precincts sitting on the district border. This second step incorporates the estimates from the first step regarding voter preferences over candidate ideology and uses the following moments to recover preferences over partisan affiliations:

$$
\mathbb{E}[w_{pt} \cdot \xi_{jpt}] = 0 \tag{12}
$$

for each demographic characteristic w_{pt} (years of schooling, race, and age). Essentially, this second step recovers the demographic heterogeneity in party preferences, accounting for the ideological preferences estimated in the first step. Since the second-step demand estimates treat the first-step parameters as known, I correct the second-step standard errors for first-step estimation uncertainty following (Murphy and Topel, 1985).

[13]Note that I still estimate voter preferences over party in the first step to account for small differences in voter demographics that might persist within precinct groups.

4.3 Demand Estimates

Table 1 reports the estimates of equation 9; corresponding estimates without within-precinct heterogeneity are also reported in Appendix Table A.6. The two tables give very similar estimates. The main parameters of interest are β_{cult} and β_{econ}, with particular focus on the heterogeneity of these dimensions across educational lines. For instance, $\beta_{edu,cult}$ captures the change in cultural preferences associated with an additional year of schooling. I begin by briefly describing the estimated coefficients and their evolution over time, before comparing the coefficients for the group of less-educated voters to those of more-educated voters. Here and throughout, less-educated voters are defined as those with education below the year-specific national median (13.65 years of schooling in 2020), while more-educated voters have education at or above the median.

The coefficients on education reveal a clear contrast between cultural and economic preferences. The estimates of $\beta_{edu,cult}$ are negative in both periods, indicating that voters with more years of schooling tend to prefer more progressive cultural positions than voters with fewer years of schooling. This relationship is similar across the two periods. By contrast, the estimates of $\beta_{edu,econ}$ are positive in both periods, indicating that more-educated voters tend to prefer more conservative economic positions than their less-educated counterparts. This economic gradient becomes substantially larger in 2012–2020, reaching a magnitude comparable to the cultural gradient. The race and ethnicity coefficients are smaller in magnitude, but suggest that, conditional on non-ideological party preferences, white voters tend to prefer more progressive positions on both cultural and economic issues than non-white voters. Finally, older voters tend to prefer more conservative cultural positions, especially in the second period, while heterogeneity by age on economic issues is small and not statistically significant in either period.

The estimates of σ_{kt}^{β} capture residual dispersion in ideological preferences that is not explained by observed demographics. The estimated standard deviation of the random coefficient on cultural positions is statistically significant in both periods and increases over time. For economic positions, the estimated dispersion is close to zero in 2000–2010 but becomes larger in 2012–2020, suggesting a rise in unobserved heterogeneity in voter preferences over the economic dimension.

Appendix Figure A.13 illustrates this heterogeneity by plotting the education gap in Democratic vote probability, separately for cultural and economic candidate positions.[14] The estimated heterogeneity in preferences aligns with the survey evidence in Kuziemko et al. (2026): less-educated voters report more conservative views on cultural issues but more progressive views on economic issues.[15]

The term α_{ijt} captures factors that affect voter i's utility from voting for party j but do not operate through the positions of the local House candidates. These factors include non-ideological party prefer-

[14]The education gap is defined as the difference in predicted Democratic vote probability between more-educated and less-educated voters. Because education, race, and age are correlated, this comparison accounts for the full demographic profile of each group, rather than changing education alone.

[15]Kuziemko et al. (2026) distinguish between preferences for predistribution policies, such as labor market interventions and trade policy, and preferences for redistribution policies, such as taxes and transfers. They show that the education gradient is strong for predistribution but more muted for redistribution.

ences, as well as the ideology of higher-level candidates, such as Senate and presidential candidates, who may exert a *coattail effect* on House vote choices (Calvert and Ferejohn, 1983). I therefore treat α_{ijt} as a control for non-ideological sources of party support, without assigning it a structural interpretation. Over time, α_{ijt} shifts significantly: more-educated voters become more likely, and white voters less likely, to support Democratic candidates. Across both periods, more-educated, white, and older voters are also more likely to turn out.

I compare my estimates with survey-based self-reported vote choice from the ANES and the CCES over the same time periods. For each period, I compute the model-based and survey-based probabilities of voting for the House Democratic candidate, conditional on voting, for 12 demographic groups defined by race, education, and age. Appendix Figure A.12 reports the results. The model-based probabilities are strongly correlated with the survey-based probabilities, both in levels (0.87 for 2000-2010 and 0.94 for 2012-2020) and in changes across the two periods (0.96). The comparison also shows that the model-based probabilities of voting are systematically more dispersed than the survey-based probabilities. The model appears to better capture the decline in Democratic vote choice among less-educated white voters, a pattern that is less visible in the survey data.

In Appendix F, I assess the robustness of the demand-side results in two ways. First, I estimate a more parsimonious multinomial logit specification without random coefficients, assuming homogeneous voters within each precinct, following Berry (1994). Second, I allow voter preferences to vary along additional candidate-level characteristics: incumbency, gender, and race and ethnicity. The estimates are generally noisier when these additional coefficients are included, as expected given that the model recovers twelve additional random coefficients, but the results remain quantitatively similar.

To summarize the demand results, Figure 8 plots education-neutral curves in the space of cultural and economic party polarization, defined as the distance between the Democratic and Republican positions on each dimension. Each curve shows the combinations of cultural and economic positions for which less-educated and more-educated voters are predicted to vote for the Democratic candidate at the same rate. The solid orange curve shows this relationship using estimated preferences from 2000-2010, while the dashed green curve uses estimated preferences from 2012-2020. Both curves are normalized to pass through the 2000 leadership position, abstracting from non-ideological changes in preferences. In the upper-left region of the policy space, Democrats perform relatively better among less-educated voters, because the parties are far apart on economic issues but relatively close on cultural issues: economic polarization pulls less-educated voters toward Democrats, while limited cultural polarization does not push them away. By contrast, in the lower-right region, Democrats perform relatively better among more-educated voters, because cultural polarization is large relative to economic polarization.

I also plot the evolution of party leadership positions between 2000 and 2020. As noted above, party leaders polarized much more on cultural issues than on economic issues over this period. In the figure, this movement shifts the parties from the education-neutral line, where Democratic candidates perform

equally well among less- and more-educated voters, toward a region where Democratic candidates perform relatively better among more-educated voters. The reason is that the disproportionate increase in cultural polarization with low economic polarization tends to deter less-educated voters from choosing Democratic candidates.

The dashed green curve shows that the change in preferences mitigated part of that loss. Its steeper slope indicates that the trade-off between cultural and economic issues has changed. Less-educated voters have become relatively more progressive on economic issues, which lowers the amount of economic polarization Democrats need, relative to cultural polarization, to retain their support. In that sense, the education-neutral curve lies closer to the final leadership position than it would have in the absence of these preference changes. Put differently, without this shift in preferences, less-educated voters would have left the Democratic Party in even larger numbers.

The next section examines the extent to which individual candidates can deviate from the leadership's position and instead align more closely with their voters' ideological preferences. This flexibility could mitigate the overall impact of the phenomenon described above, potentially reducing the degree of voter realignment driven by leadership shifts.

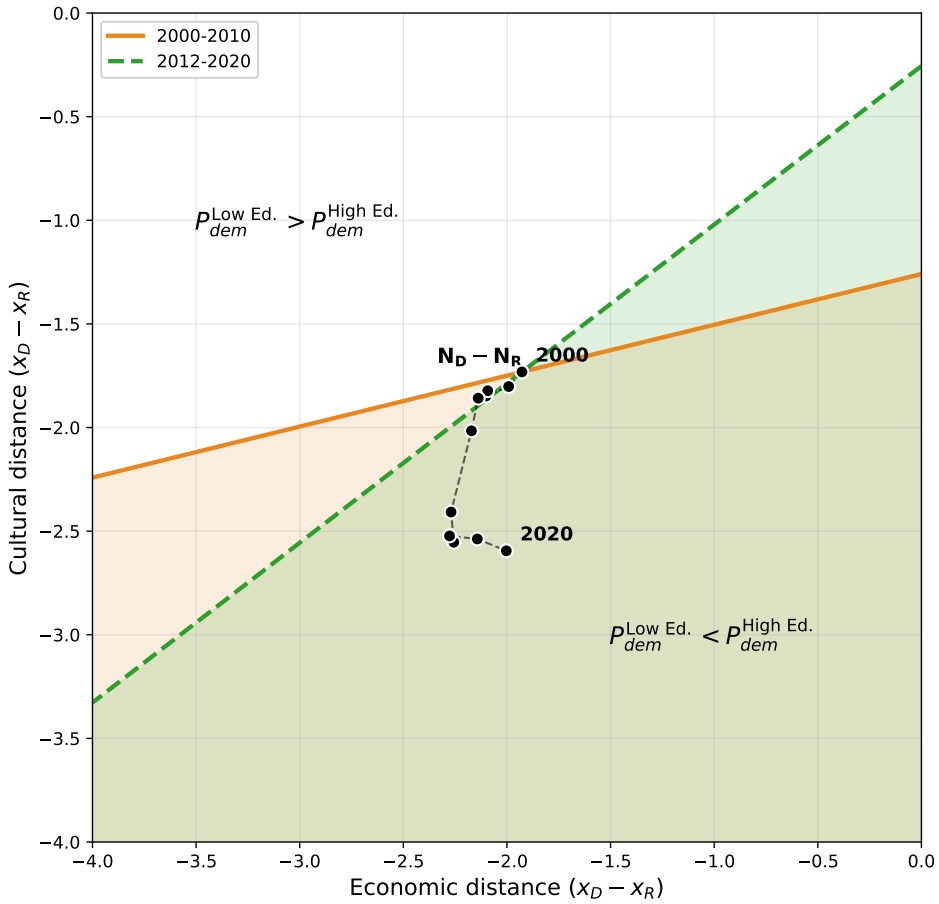


Figure 8: Education Gap in Democratic Vote Probability by Levels of Polarization

Notes: The figure plots the education-neutral curve, that is, the combinations of cultural and economic party polarization for which less-educated and more-educated voters are predicted to vote for the Democratic candidate at the same rate. The x-axis shows polarization on economic issues, while the y-axis shows polarization on cultural issues, in each case measured as the distance between the Democratic and Republican positions. The solid orange curve uses estimated preferences from 2000-2010, while the dashed green curve uses estimated preferences from 2012-2020. Combinations above and to the left of a curve correspond to positions for which less-educated voters are relatively more likely than more-educated voters to support Democratic candidates; combinations below and to the right correspond to positions for which more-educated voters are relatively more likely to support Democratic candidates. The dashed black path traces the evolution of party leadership positions between 2000 and 2020. Appendix Figure A.13 shows the education gap in vote probability, separately for each dimension.

Coefficients	Parameters	Estimates	Standard Errors
First Period: 2000-2010			
Cult	β_{cult}	0.2762	(0.0722)
Econ	β_{econ}	-0.0324	(0.0674)
Dem intercept	α_D	1.1942	(0.3468)
Baseline Yrs. Schooling	α_{edu}	0.1852	(0.0148)
Dem × Yrs. Schooling	$\alpha_{D,\text{edu}}$	0.0895	(0.0186)
Baseline White	α_W	3.6524	(0.1450)
Dem × White	$\alpha_{D,W}$	-3.7737	(0.1576)
Baseline Age	α_{age}	0.0232	(0.0034)
Dem × Age	$\alpha_{D,\text{age}}$	0.0206	(0.0041)
Cult × Yrs. Schooling	$\beta_{\text{edu,cult}}$	-0.0190	(0.0045)
Cult × White	$\beta_{W,\text{cult}}$	-0.0611	(0.0311)
Cult × Age	$\beta_{\text{age,cult}}$	0.0009	(0.0015)
Econ × Yrs. Schooling	$\beta_{\text{edu,econ}}$	0.0060	(0.0040)
Econ × White	$\beta_{W,\text{econ}}$	-0.0181	(0.0270)
Econ × Age	$\beta_{\text{age,econ}}$	-0.0011	(0.0014)
Unobserved cultural heterogeneity	$\sigma^{\beta}_{\text{cult}}$	0.0477	(0.0136)
Unobserved economic heterogeneity	$\sigma^{\beta}_{\text{econ}}$	0.0017	(0.0076)
Second Period: 2012-2020			
Cult	β_{cult}	0.1663	(0.0613)
Econ	β_{econ}	-0.2391	(0.0633)
Dem intercept	α_D	-0.8827	(0.3015)
Baseline Yrs. Schooling	α_{edu}	0.1473	(0.0183)
Dem × Yrs. Schooling	$\alpha_{D,\text{edu}}$	0.2990	(0.0184)
Baseline White	α_W	3.9651	(0.1501)
Dem × White	$\alpha_{D,W}$	-4.4093	(0.1486)
Baseline Age	α_{age}	0.0437	(0.0031)
Dem × Age	$\alpha_{D,\text{age}}$	0.0073	(0.0026)
Cult × Yrs. Schooling	$\beta_{\text{edu,cult}}$	-0.0185	(0.0037)
Cult × White	$\beta_{W,\text{cult}}$	-0.0430	(0.0211)
Cult × Age	$\beta_{\text{age,cult}}$	0.0034	(0.0010)
Econ × Yrs. Schooling	$\beta_{\text{edu,econ}}$	0.0159	(0.0035)
Econ × White	$\beta_{W,\text{econ}}$	-0.0565	(0.0245)
Econ × Age	$\beta_{\text{age,econ}}$	0.0006	(0.0011)
Unobserved cultural heterogeneity	$\sigma^{\beta}_{\text{cult}}$	0.0581	(0.0126)
Unobserved economic heterogeneity	$\sigma^{\beta}_{\text{econ}}$	0.0147	(0.0086)

Table 1: Estimated demand-side parameters with demographic heterogeneity

Notes: The table shows the estimated coefficients from Equation 9, estimated by GMM with precinct-by-party and precinct-group-by-party-by-election fixed effects. Standard errors are clustered by congressional district by election. Standard errors from the second step are adjusted for the first step uncertainty. Results without within-precinct heterogeneity (Berry, 1994) are reported in Appendix Table A.6. Appendix Table A.4 provides descriptive statistics on voter demographics.

5 Estimating candidates' ideological positioning

5.1 Model

After estimating the demand-side parameters, which capture the distribution of voter ideological preferences across congressional districts, I model candidates' positioning decisions. Specifically, I estimate a standard candidate competition model (e.g., Wittman (1983)) in which candidates balance the need to cater to their constituents' preferences with the pressure to adhere to their party's platform. This framework allows me to identify the factors influencing candidates' strategic decisions in offering policy positions that may deviate from or align with the party line.

I define each candidate's objective function as Π_{jdt}, where candidates seek to maximize their probability of being elected, given that they derive a positive rent $Q > 0$ from holding office, while simultaneously minimizing their deviation from the national party platform. This framework captures the trade-off candidates face between aligning with their local voters' preferences and adhering to the national party line. I write

$$\Pi_{jdt}(\mathbf{x_{jdt}}) = \lambda_{jt} \underbrace{P_{jdt}(\mathbf{x_{jdt}}, \mathbf{x_{-jdt}})}_{\text{probability of winning}} Q - \underbrace{||\mathbf{x_{jdt}} - \mathbf{N_{jt}}||^2}_{\substack{\text{cost of deviation from} \\ \text{national party platform}}}, \tag{13}$$

where the parameter λ_{jt} captures the weight candidates place on electoral incentives relative to alignment with the national party platform. $\frac{1}{\lambda_{jt}}$ can be interpreted as the relative cost of deviating from the party line.

There is an aggregate shock ζ_{jdt} that introduces uncertainty into candidates' probability of winning (Persson and Tabellini, 2002). Assuming ζ_{jdt} follows a uniform distribution:[16] $\zeta_{jdt} \sim U(-\frac{1}{\phi}, \frac{1}{\phi})$, I can write

$$\begin{aligned} P_{jdt}(\mathbf{x_{jdt}}, \mathbf{x_{-jdt}}) &= \mathbb{P}(\widetilde{s}_{jdt}(\mathbf{x_{jdt}}, \mathbf{x_{-jdt}}) + \zeta_{jdt} \geq 0.5) \\ &= 1 - \mathbb{P}(\zeta_{jdt} \leq 0.5 - \widetilde{s}_{jdt}(\mathbf{x_{jdt}}, \mathbf{x_{-jdt}})) \\ &= \frac{\phi}{2}\widetilde{s}_{jdt}(\mathbf{x_{jdt}}, \mathbf{x_{-jdt}}) + \frac{1}{2} - \frac{\phi}{4}. \end{aligned}$$

where $\widetilde{s}_{jdt}(\mathbf{x_{jdt}}, \mathbf{x_{-jdt}})$ is the vote share conditional on turnout.

Taking the first-order conditions with respect to each ideal point dimension gives the following equilibrium conditions:

$$(x_{jdkt} - N_{jkt}) = \widetilde{\lambda}_{jt} \frac{\partial \widetilde{s}_{jdt}(\mathbf{x_{jdt}}, \mathbf{x_{-jdt}})}{\partial x_{jdkt}} + \eta_{jdkt}, \tag{14}$$

[16] While not an uncommon assumption in political economy (see e.g., Acemoglu et al. (2013)), uniform shocks lead to a linear relationship between vote shares and candidates' objective functions. Supplemental Appendix G therefore shows that estimates of λ under an alternative distributional assumption for the shock, using a Logistic distribution, give very similar results.

on each dimension k, and where $\frac{\partial \widetilde{s}_{jdt}(\mathbf{x_{jdt}}, \mathbf{x_{-jdt}})}{\partial x_{jdkt}}$ is the derivative of the demand function with respect to candidate position on dimension k, conditional on voting, recovered from the demand estimation, $\widetilde{\lambda}_{jt} = \frac{\lambda_{jt}\phi Q}{4}$, and η_{jdkt} captures deviations from the first-order condition, including candidate-level unobservables, measurement error in candidate positions, or any other residual factors affecting observed candidate positioning. I only recover the strength of party discipline relative to candidates' rent from being elected (Q) and the support of the aggregate shock (ϕ). Intuitively, the smaller the support of the shock and the greater the rent, the stronger the actual level of party discipline will be for a given parameter $\widetilde{\lambda}$.

5.2 Estimation and Identification

Estimation of the supply-side parameters is conducted conditional on the estimated demand system. Given the demand estimates, I compute the model-implied derivative of vote shares with respect to candidates' policy positions. Equation (14) is linear in the supply-side parameter $\widetilde{\lambda}_{jt}$, allowing me to recover the strength of party discipline from the relationship between candidates' deviations from party leadership and the predicted slope of voter demand in their district. I allow for a nonzero mean of the unobserved ideological component η_{jdkt} to account for systematic deviations of candidates from the leadership. A remaining concern is that the demand derivative is evaluated at observed candidate positions, which are themselves mechanically correlated with η_{jdkt}. To address this concern, I instrument for the actual demand derivative using the derivative evaluated at the median candidate position from each party. This instrument preserves variation in the slope of demand coming from district-level voter composition, while removing variation mechanically induced by the candidate's own realized position.

5.3 Supply Estimates

Table 2 presents the estimated parameters for each period and party, where a lower λ reflects a greater electoral cost for candidates deviating from the party line. I report results from three different specifications: the first corresponds to the specification described in equation 14, the second controls for party by dimension by year fixed effects, and the third excludes party leaders from the estimation. All three specifications produce similar estimates. In Appendix G, I estimate an alternative model in which the aggregate supply shock follows a Logistic rather than a uniform distribution, which gives very similar estimates.

The estimates of λ are broadly comparable across the two parties within each period. In 2000–2010, candidates placed substantial weight on local electoral incentives when choosing their positions. The magnitude of the coefficient implies that, in a district where moving one standard deviation to the left would locally increase the candidate's vote share by one percentage point, candidates deviate from the national party line by approximately one-quarter of a standard deviation in that direction. Between the two periods, this weight declines sharply, implying a substantial increase in party discipline. In 2012–2020,

	$\widetilde{\lambda}$		
Rep 2000-2010	22.2550	22.8928	22.9202
	(4.1075)	(4.0966)	(4.1017)
	[9.3378]	[9.4674]	[9.4990]
Rep 2012-2020	-3.9669	-0.6326	-0.5328
	(3.7400)	(3.7973)	(3.8118)
	[6.9263]	[6.8587]	[6.9069]
Dem 2000-2010	22.6623	22.6886	21.9412
	(5.5916)	(5.6094)	(5.6190)
	[17.8059]	[18.0670]	[17.8282]
Dem 2012-2020	-5.8159	-3.7325	-3.3970
	(3.0147)	(3.0566)	(3.0629)
	[6.4151]	[6.3011]	[6.2183]
Party × dimension × year FE	No	Yes	Yes
Sample	All	All	Excluding Party Leaders

Table 2: Estimated supply parameters

Notes: The table shows the supply coefficients from equation 14. All columns use a weighted average measure of party leadership. The second column includes party by dimension by year fixed effects. The third column excludes main party leaders from the analysis. Conventional standard errors are reported in parentheses, and standard errors that adjust for demand-side estimation uncertainty (Murphy and Topel, 1985) are reported in brackets below.

the estimated weight on local demand is no longer distinguishable from zero.[17] This pattern is consistent with previous evidence of rising party discipline in Congress, such as Canen et al. (2020) and Canen et al. (2026), as well as evidence based on campaign contributions from party PACs (Cox and Shapiro, 2024).

The standard-error adjustment for demand-side estimation uncertainty, following Murphy and Topel (1985), provides a more conservative assessment of the precision of the supply estimates. This adjustment is important because the supply-side estimates depend directly on generated demand-side objects, in particular the estimated demand derivatives. As expected, accounting for this first-step uncertainty increases the standard errors, especially for the Democratic estimate in the first period, which is no longer statistically distinguishable from zero with the adjusted standard errors.

For a candidate in a given district, stronger party discipline can generate electoral losses when the national party platform diverges from local voter preferences. In practice, this also leads to greater uniformity in candidate positions within each party across the country, as candidates remain closer to the national platform regardless of district-specific conditions.

It is important to note that the interpretation of party discipline in this paper is broad and not tied to a single micro-founded mechanism. I do not observe the specific constraints that keep candidates

[17] Note that the estimates from the logit shock specification in Appendix G yield positive, small, and more precisely estimated coefficients for the second period.

aligned with the national party platform. Instead, I interpret systematic limits on candidates' ability to adjust their positions to local voter preferences as reflecting party discipline. These limits could arise from several mechanisms, including pressure from party leadership, career concerns, changes in primary voter preferences or primary structure, or selection of politicians into candidacy who are more ideologically aligned with the national party. Overall, the objective of the supply model is to capture the trade-off between local electoral incentives and national party alignment, with party discipline measuring the strength of national-party constraints relative to local demand, without distinguishing among the channels through which these constraints operate.

6 Political Realignment: Demand vs. Supply Factors

The previous two sections estimate the factors influencing voter choices based on candidate positions and candidates' positioning decisions in response to district composition and party dynamics. With these estimates in hand, I simulate counterfactual equilibrium outcomes to explore the contribution of each demand-side and supply-side factor to the observed changes in voting behavior. Each counterfactual scenario is characterized by a set of parameters that gives a unique equilibrium defined by $(\mathbf{x}^*, \mathbf{s}^*) = f\left(F_{pt}\left(\mathbf{w}\right), \boldsymbol{\beta_t}, \mathbf{N_t}, \boldsymbol{\lambda_t} \middle| \mathbf{m_t}, \boldsymbol{\alpha_t}, \boldsymbol{\xi_t}, \boldsymbol{\eta_t}\right)$ where $\mathbf{x}^*, \mathbf{s}^*$ are vectors of candidate positions and vote probabilities, $F_{pt}(\mathbf{w})$ is the distribution of demographics in each precinct p, $\boldsymbol{\beta_t}$ captures voter ideological preferences, $\mathbf{N_t}$ denotes national party leaders' positions, $\boldsymbol{\lambda_t}$ measures the weight given to electoral incentives in candidates' choice of positions, $\mathbf{m_t} = (m_{p,d})_{\forall p,d}$ is a redistricting scenario, that is, a map of precincts to districts, $\boldsymbol{\alpha_t}$ accounts for voter partisan preferences, $\boldsymbol{\xi_t}$ is a vector of precinct-level taste shocks, and $\boldsymbol{\eta_t}$ is a vector of candidate-positioning residuals.

Under each scenario, I determine new equilibrium candidate positions as the solution to a fixed point algorithm, where each candidate plays their best response to the other's strategy. The fixed point is solved using an imitation game algorithm (McLennan and Tourky, 2005; Batista et al., 2024). These counterfactual candidate positions are then used as inputs in the demand system to generate counterfactual electoral outcomes. I assess the impact of each factor by comparing the realized outcomes under both its initial and final values. Since the effect of each factor depends on the values of the others, I calculate the average contribution for each factor across all possible permutations of the remaining factors, similar to a Shapley value decomposition (Shapley, 1953), following the approach of Guriev et al. (2023). Appendix H details the computation of each factor's contribution.

I assess how changes in each factor contributed to the education gap in Democratic voting. Figure 9 presents the relative contributions of demand-side and supply-side factors to this gap, defined as the difference between the probability that more-educated voters vote for Democratic candidates and the probability that less-educated voters do so. Appendix Figure A.14 shows the impact of each factor on vote probabilities for each group separately. As shown in Section 3, this gap increased substantially

between the two periods. The first two bars show that both demand-side factors reduced the education gap. Demographic changes benefited Democratic candidates among both more-educated and less-educated voters, as the population became more educated and more racially diverse, but these gains were larger among less-educated voters. Changes in voters' ideological preferences, $\boldsymbol{\beta}$, also reduced the gap. In particular, less-educated voters became more progressive on economic issues, making Democratic candidates relatively more attractive to them than they would have been absent these preference changes.

In contrast, supply-side factors pushed the education gap in the opposite direction. The increasing polarization of party leadership on cultural issues moved less-educated voters toward Republican candidates and more-educated voters toward Democratic candidates, increasing the education gap. This effect was only partially offset by changes in leadership positions on economic issues. The rise in party discipline had little net effect on the education gap: because discipline increased in both parties, it limited local adaptation by both Democratic and Republican candidates and changed Democratic vote probabilities by similar amounts among more-educated and less-educated voters.

The underlying dynamic can be summarized as follows: while parties were equally polarized on economic and cultural issues in the early 2000s, they have since diverged much more sharply on cultural issues. Less-educated voters, who tend to prefer progressive economic policies but conservative cultural ones, now derive less utility from supporting Democratic candidates and have increasingly moved toward Republicans. Conversely, more-educated voters derive greater utility from Democratic candidates as party polarization has become increasingly concentrated on cultural issues. Demand-side changes have partly offset these supply-side forces: less-educated voters have become increasingly progressive on economic issues, bringing them closer to Democratic candidates than they would have been absent these preference changes.

As mentioned earlier, the framework developed in this paper focuses on ideological variation operating through House candidates' positions and therefore does not fully explain changes in vote choices between the two periods. A significant portion of these changes is captured by α_{ijt}, which reflects voters' general utility from supporting party j, independent of the positions of local House candidates. This component may reflect non-ideological party preferences, as well as the ideology of higher-level candidates, such as presidential and Senate candidates, who may exert a *coattail effect* on House candidate choices (Calvert and Ferejohn, 1983). Although the impact of presidential, senatorial, and party leaders' positions on House vote shares cannot be separately identified within this framework, these factors are likely to operate through similar ideological channels. To the extent that they do, the estimates presented here may be a lower bound of the overall effect of party leadership's ideological shifts.

It is important to emphasize that realignment is not driven simply by rising party polarization, but by the changing balance of polarization across issue dimensions. In particular, parties have diverged much more on cultural issues than on economic issues. Had economic polarization increased alongside cultural polarization, the observed realignment by education would have been less pronounced.

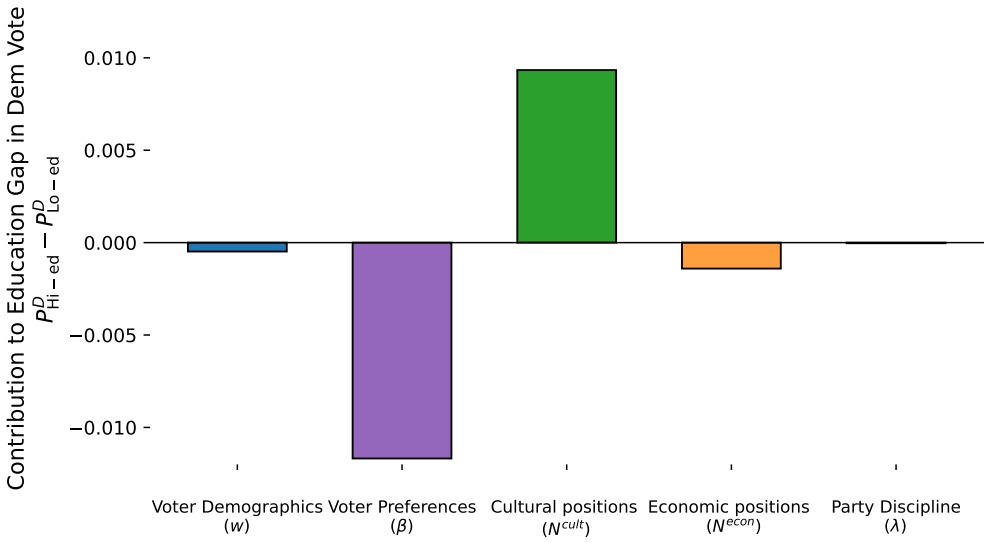


Figure 9: Equilibrium Contributions to the Education Gap in Democratic Voting

Notes: The figure shows the contributions of changes in demand-side and supply-side factors to the education gap in Democratic voting probability, defined as the difference between the probability that voters with above-median education vote for the Democratic candidate and the probability that voters with below-median education do so. For each voter, I compute the Shapley-value contribution of each factor to the change in their predicted probability of voting for the House Democratic candidate, as described in Appendix H. The decomposition is computed in equilibrium, so each contribution incorporates both the direct effect of the corresponding factor and the indirect effect operating through endogenous candidate positioning. The overall education gap in Democratic voting increased by 7.4 percentage points between the two periods. The nuisance factors $\boldsymbol{\alpha}_t$ and $\mathbf{m_t}$ which are included in the Shapley decomposition but are not assigned a structural interpretation, are omitted from the figure. Appendix Figure A.14 shows the change in vote probabilities separately for each education group.

7 Conclusion

This paper studies whether recent political realignment is driven mainly by changes in voter preferences or by changes in party and candidate positioning. Distinguishing these mechanisms is difficult from survey data or election results alone, because voters choose among candidates whose positions are themselves strategic, potentially endogenous to voter preferences, and often difficult to observe. The issue is central to recent U.S. politics: parties have moved further apart (Hare and Poole, 2014; Gentzkow et al., 2019), while voters have shifted their partisan allegiances over the past two decades (Kitschelt and Rehm, 2019; Gethin et al., 2022). I connect these two facts by building and estimating a multidimensional political equilibrium model of U.S. House elections, combining new measures of candidate ideology with precinct-level estimates of voter demand and a model of candidate positioning.

I show that less-educated voters prefer more conservative cultural policies but more progressive economic ones, with their preference for progressive economic policies increasing over time. In parallel, political parties have polarized much more on cultural issues than on economic ones, and rising party discipline has constrained local candidates' ability to adapt to local conditions. Combining these estimates in counterfactual scenarios, I find that most of the increase in the education gap in Democratic voting can be attributed to supply-side factors. In contrast, concurrent ideological changes in voter preferences among less-educated voters have favored Democratic candidates, suggesting that political realignment would have been even more pronounced absent these preference changes.

These findings speak to the question of why democratic politics has not slowed rising inequality (Roemer, 1998; Bonica et al., 2013; Hacker and Pierson, 2020). This paper shows that the decline of class voting has not been driven by less-educated voters abandoning progressive economic policy. On the contrary, their support for such positions has increased over time. Instead, as parties have polarized more sharply on cultural than on economic issues, partisan choice has become more strongly shaped by cultural positioning and less aligned with class-based economic preferences. In this sense, cultural polarization has weakened the ability of electoral competition to translate class-based economic preferences into inequality-reducing policy.

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A Notation

Symbol	Description
p	index of precinct (block-group)
d	index of congressional district ($d(p)$ district of precinct p)
t	index of election year
j	index of party
k	index of policy dimension
i	index of voter
$g(p)$	precinct-group
x_{jdkt}	position of candidate from party j in district d on dimension k in election t
ξ_{jpt}	precinct p's unobserved taste shock for party j at election t
ξ_{jp}	precinct p's constant unobserved taste shock for party j
$\xi_{jg(p)t}$	precinct-group $g(p)$ unobserved taste shock for candidates for party j at election t
$\widetilde{\xi}_{jpt}$	precinct p's residual unobserved taste shock for party j at election t
ϵ_{ijpt}	Voter i's unobserved taste shock for party j in precinct p at election t
β_{ikt}	Voter i's preference over candidate's positions on dimension k at election t
α_{ijt}	Voter i's "non-ideological" partisanship in support of party j at election t
σ^{β}_{kt}	standard deviation of random coefficients on dimension k at time t
$\mathbf{w}_{it}$ (resp. $\mathbf{w}_{pt}$)	Demographics of voter i (resp. precinct p) at election t
$F_{pt}(\mathbf{w})$	Distribution of demographics in precinct p at election t
s_{ijpt}	Voter i's probability of voting for party j in precinct p at election t
s_{jpt}	Party j's share in precinct p at election t
$\widetilde{s}_{jpt}$	Party j's vote share in precinct p at election t, conditional on turnout
S_{jpt}	Party j's observed share in precinct p at election t
$\widetilde{S}_{jpt}$	Party j's observed vote share in precinct p at election t, conditional on turnout
ν_{it}	Unobserved voter i heterogeneity in preferences at time t
Π_{jdt}	Candidate from party j in district d's objective function at time t
λ_{jt}	Weight on electoral incentives in election t for party j
Q	Candidate's rent from being elected
N_{jkt}	Position of party j's leadership on dimension k at time t
$\mathbf{m}$	Mapping of precincts p to districts d: $\mathbf{m} = (m_{pd})_{\forall p,d}$

B Additional Tables and Figures not shown in the main paper

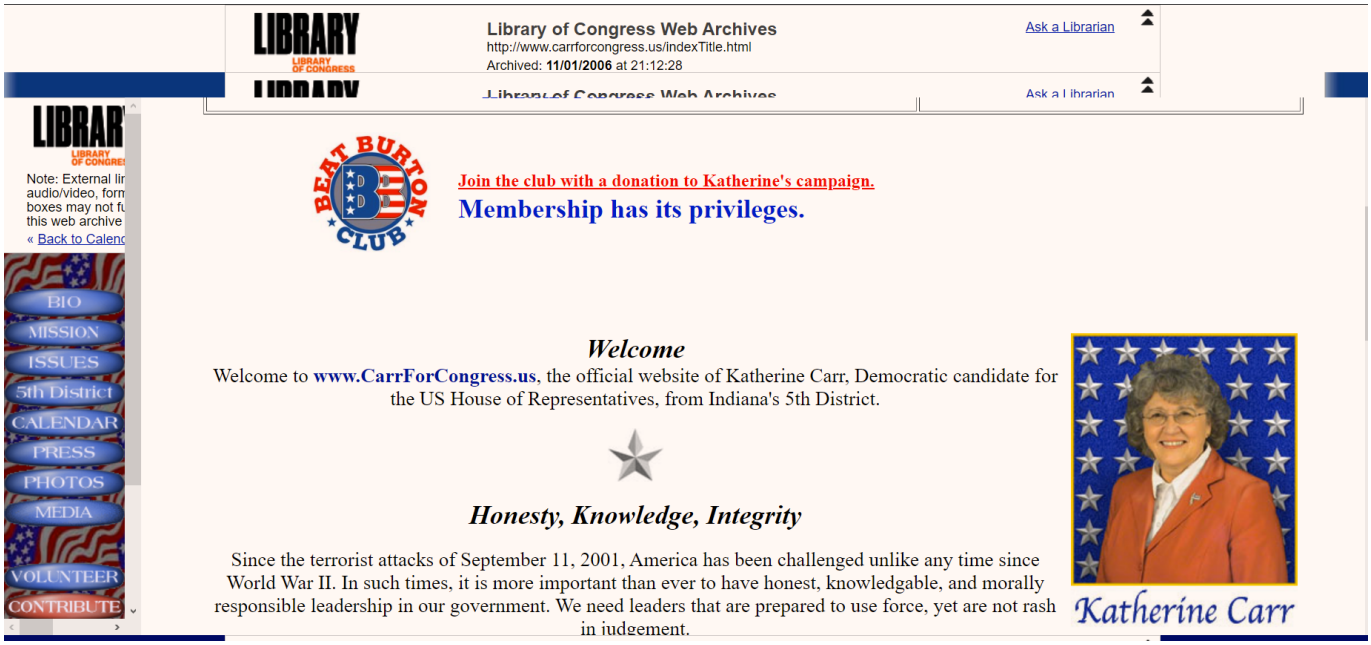


Figure A.1: Example of a website for a U.S. House candidate in Indiana in 2006

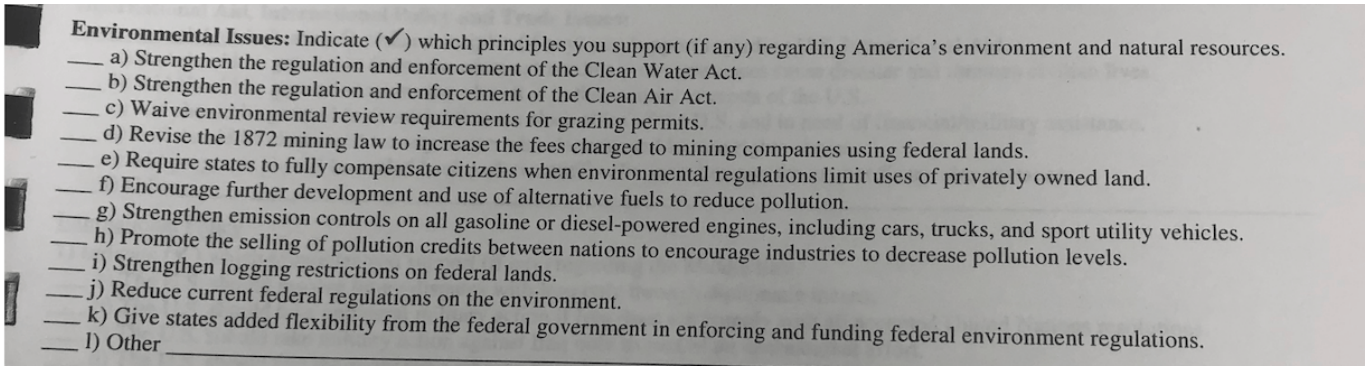


Figure A.2: Example of candidate survey (environment section in 2002)

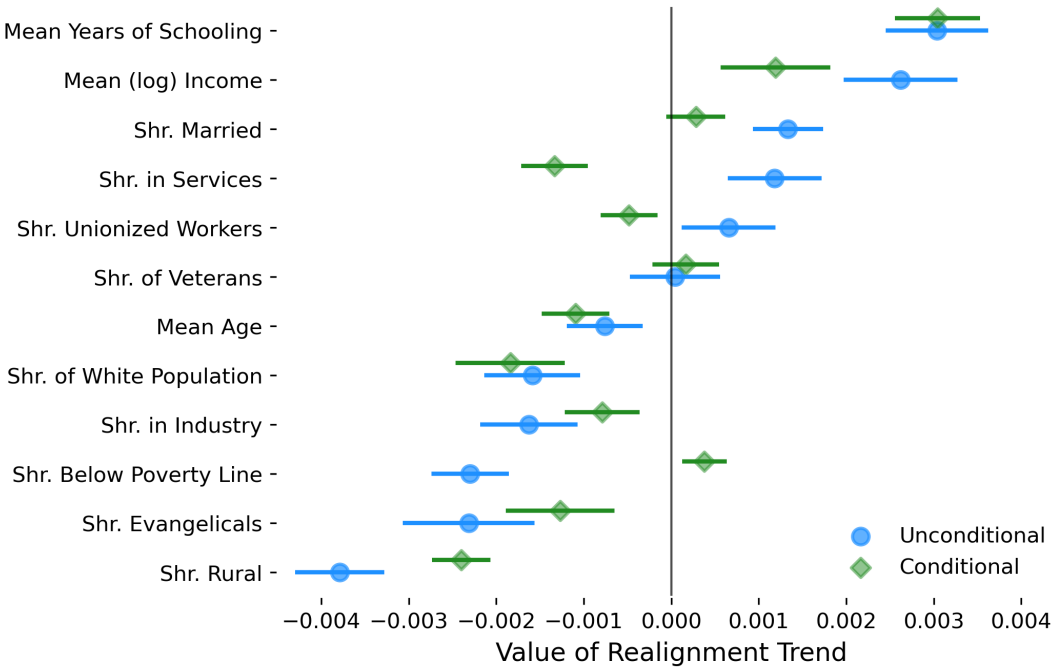


Figure A.3: Observed Trends in Political Realignment - Unconditional Estimates

Notes: The figure shows, for each demographic dimension m, the coefficients β^m from the following linear regression: $\widetilde{S}_{Dpt} = \beta_m\, w_{m,pt}\, t \;+\; \gamma_m\, w_{m,pt} \;+\; \mu_t + \varepsilon_{pt}$ where $\widetilde{S}_{Dpt}$ is the House Democratic vote share in precinct p at time t, conditional on turnout, $w_{m,pt}$ captures demographic dimension m, and μ_t are election fixed-effects. Positive coefficients indicate realignment toward the Democratic Party while negative coefficients indicate realignment toward the Republican Party. The unconditional regression (shown with a round blue marker) tests for a trend separately for each variable, while the conditional coefficients are as in figure 2. The bars around each marker show the 95% confidence intervals with standard errors clustered at the congressional-district-by-year level.

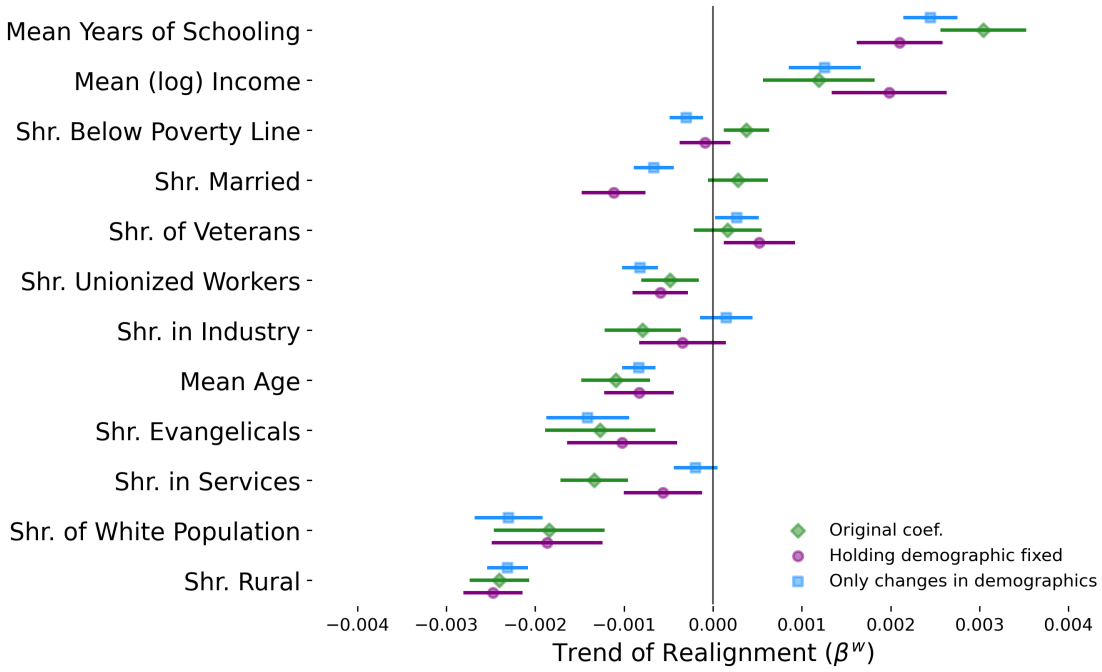


Figure A.4: Robustness of Observed Trends in Political Realignment

Notes: As in Figure 2, the figure shows, for each demographic dimension m, the coefficients β^m from the following linear regression: $\widetilde{S}_{Dpt} = \sum_{m=1}^{M} \beta_m \, w_{m,pt} \, t \; + \; \sum_{m=1}^{M} \gamma_m \, w_{m,pt} \; + \; \mu_t + \varepsilon_{pt}$ where $\widetilde{S}_{Dpt}$ is the House Democratic vote share in precinct p at time t, conditional on turnout, $w_{m,pt}$ captures demographic dimension m, and μ_t are election fixed-effects. Positive coefficients indicate realignment toward the Democratic Party while negative coefficients indicate realignment toward the Republican Party. The bars around each marker show the 95% confidence intervals with standard errors clustered at the congressional-district-by-year level. The green diamond markers show the original coefficients, as in Figure 2, the purple circle markers show the coefficients from a specification using the precinct demographics from 2000, therefore accounting for any sorting. The blue square markers show the coefficients from a specification with precinct fixed effects, therefore using only changes over time.

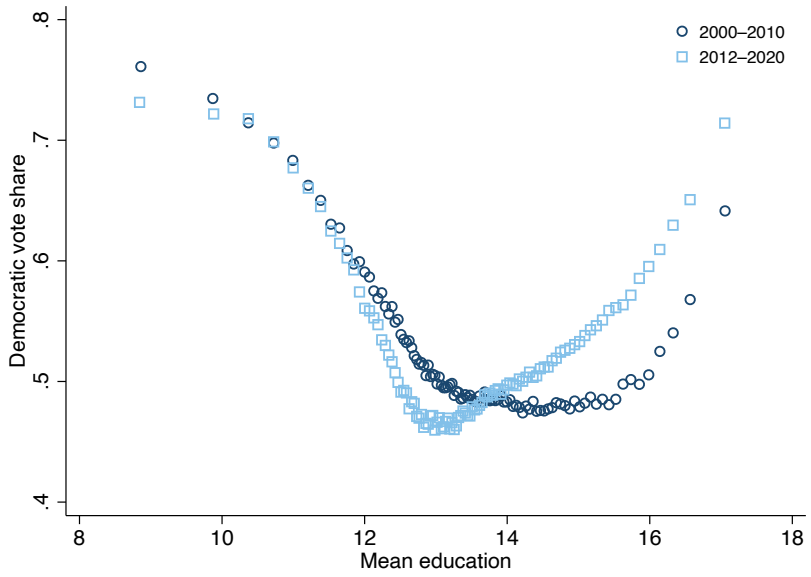


Figure A.5: Relationship between education and Democratic vote.

Notes: The figure plots average Democratic vote share against average education across 100 bins of the precinct-level education distribution, separately for 2000–2010 and 2012–2020. Education is measured as the average years of schooling in the precinct. Both variables are residualized with respect to year fixed effects, so the figure abstracts from differences in levels across years.

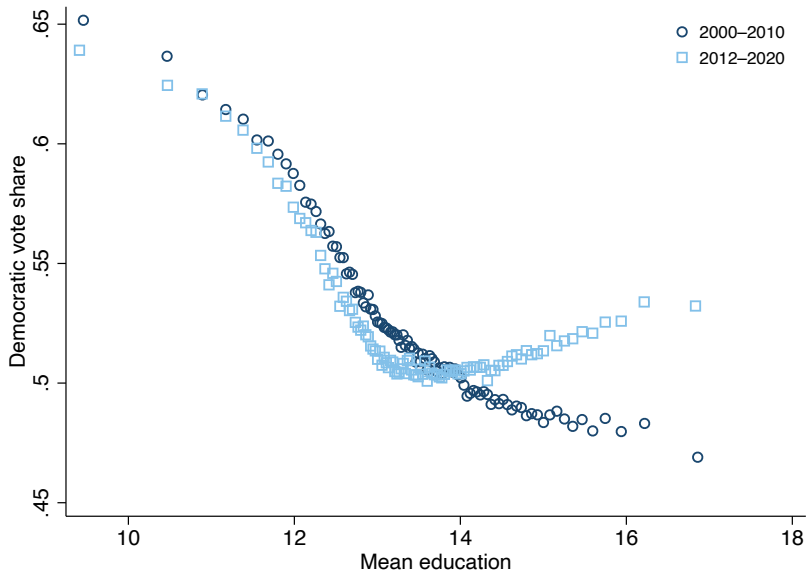
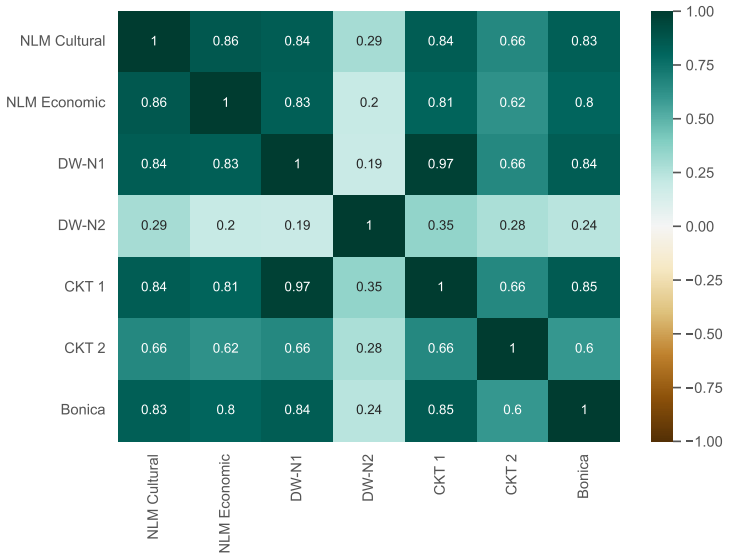
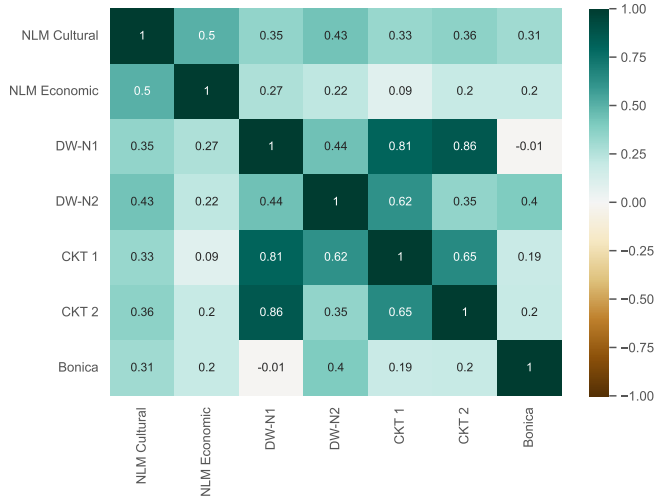


Figure A.6: Relationship between education and Democratic vote within congressional districts.

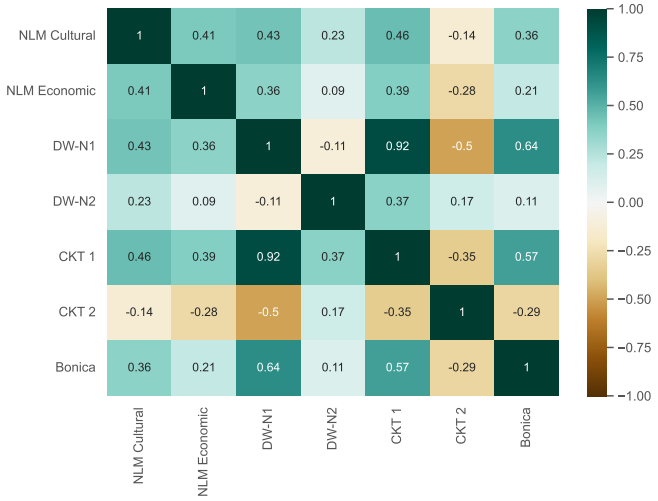
Notes: The figure plots average Democratic vote share against average education across 100 bins of the precinct-level education distribution, within congressional district, separately for 2000–2010 and 2012–2020. Education is measured as the average years of schooling in the precinct.



(a) All Sample



(b) Democrats only



(c) Republicans only

Figure A.7: Comparison of the paper's measures with common measures of ideology

Notes: The first panel shows the pairwise Spearman (rank) correlation between measures computed for this paper (denoted as NLM) and other commonly used measures of ideology. The correlations with DW-Nominate first and second dimensions (DW-N1 and DW-N2) are only for House and Senate winners. The pairwise correlations with Canen, Kendall, and Trebbi (2026) use only Senate election winners. The pairwise correlations with Bonica (2014) use all candidates. The second and third panels show the within-party correlations for Democrats and Republicans, respectively.

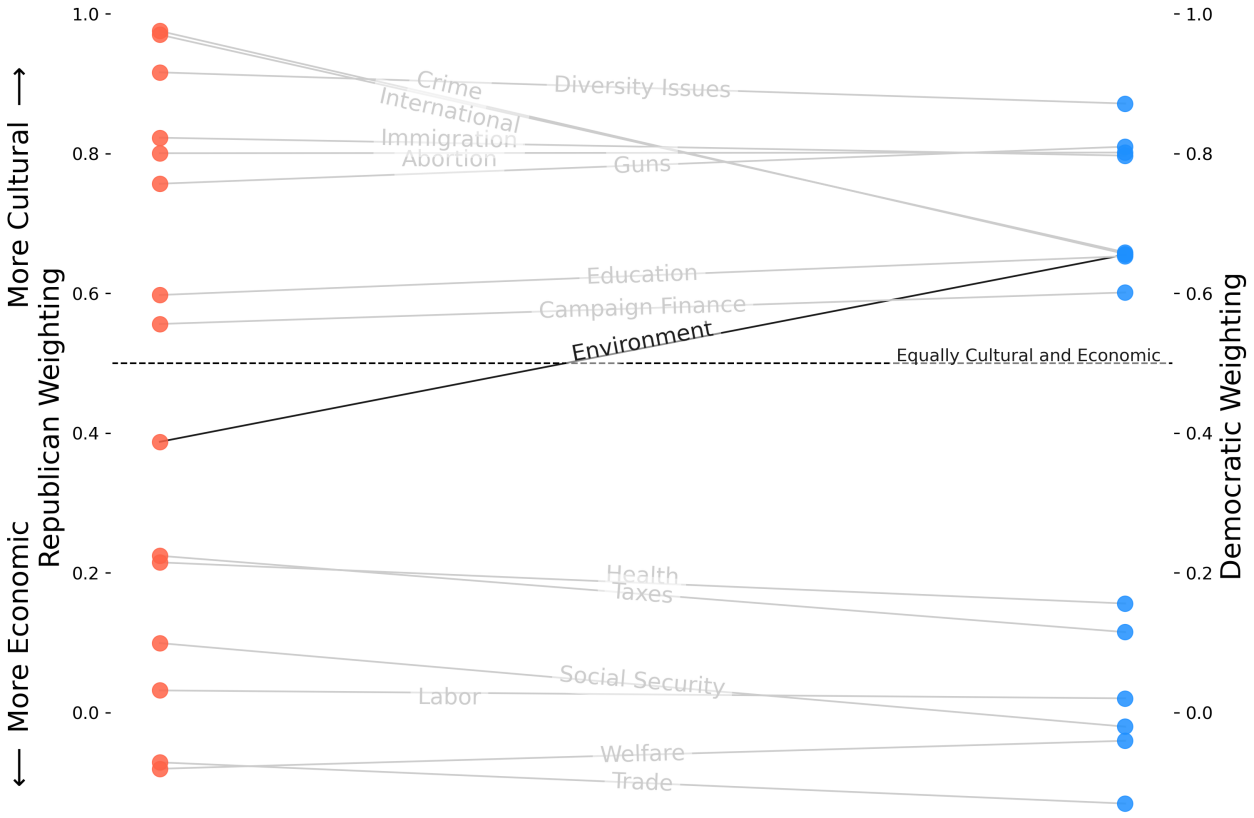
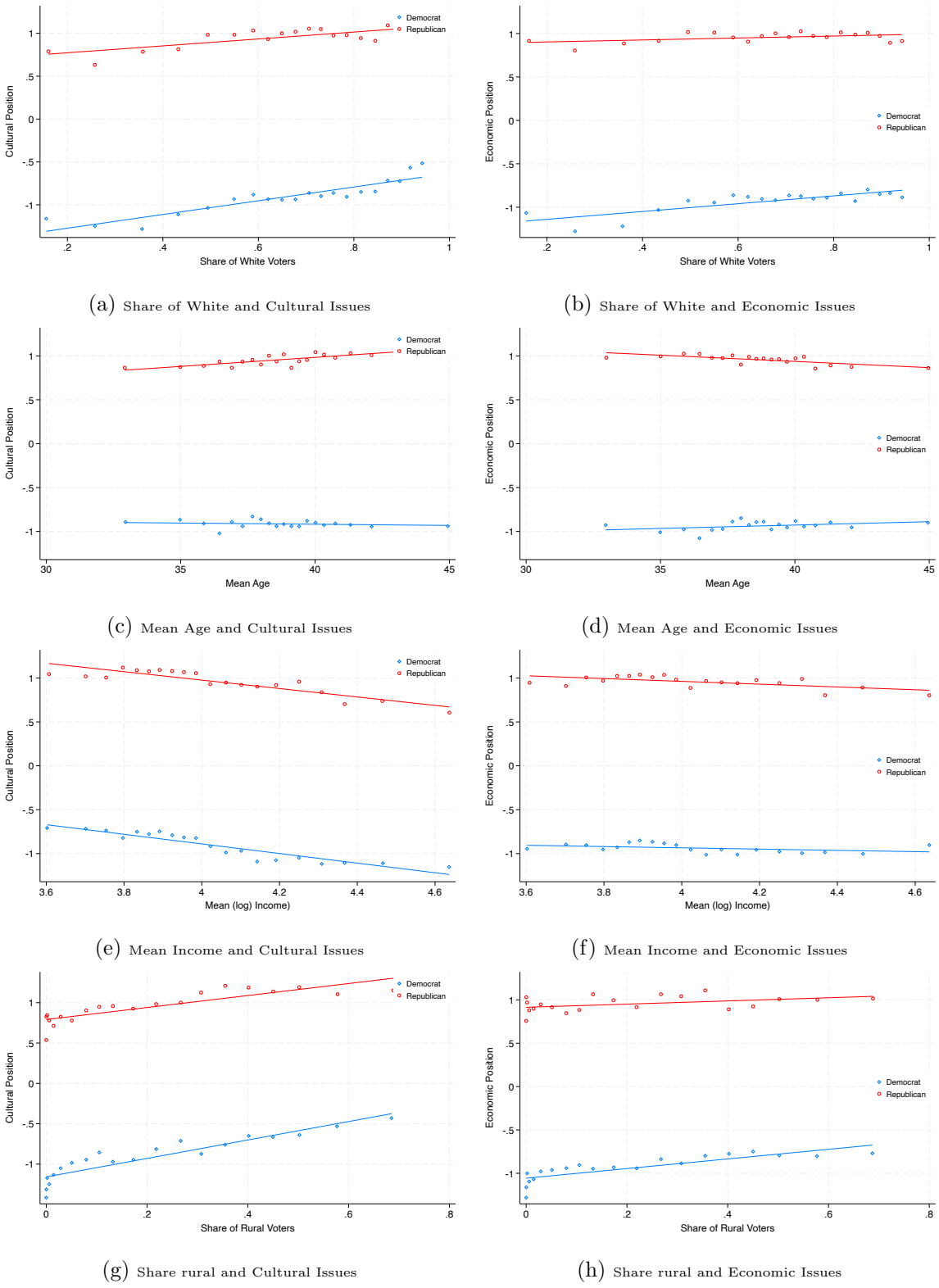


Figure A.8: Party Cultural Weights on each Topic

Notes: For each of the 15 topics, I re-estimate the ideal point model to get a position specifically on that topic and positions on the cultural and economic dimensions, excluding that topic. I then project the topic ideal points on the cultural and economic dimensions by regressing the topic-specific ideal point on the economic and cultural ideal points: $x_{j,k} = \gamma_0 + \gamma_{k,cult} x_{j,cult} + \gamma_{k,econ} x_{j,econ} + \zeta_{jk}$, for each topic k, separately for each party. I plot the relative weighting coefficients $\rho_k = \frac{\gamma_{k,cult}}{\gamma_{k,cult} + \gamma_{k,econ}}$.



(a) Share of White and Cultural Issues

(b) Share of White and Economic Issues

(c) Mean Age and Cultural Issues

(d) Mean Age and Economic Issues

(e) Mean Income and Cultural Issues

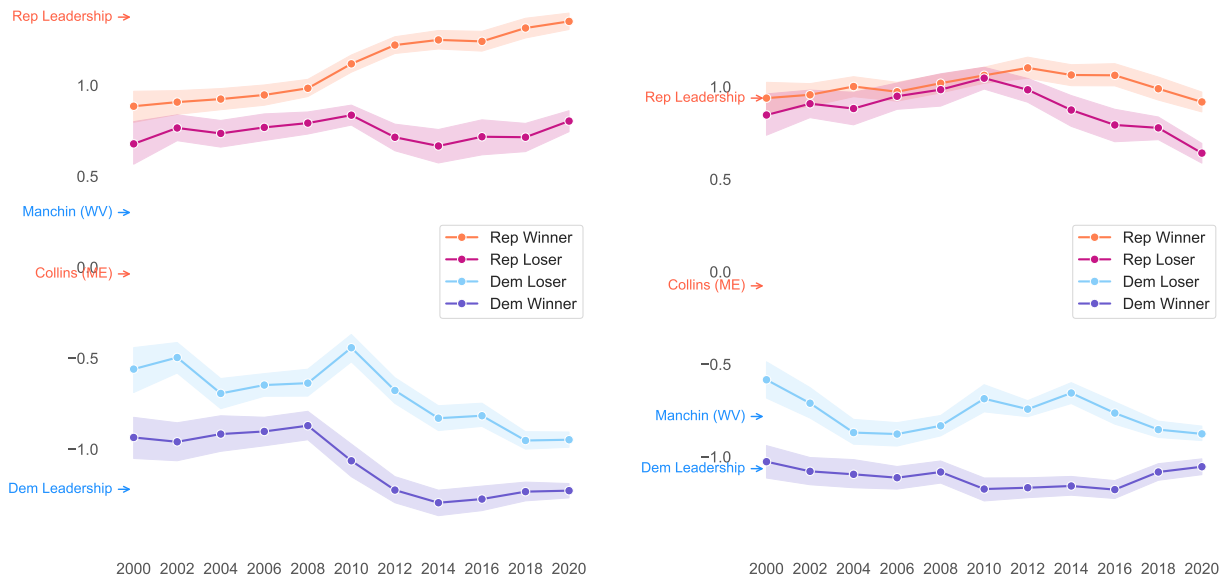
(f) Mean Income and Economic Issues

(g) Share rural and Cultural Issues

(h) Share rural and Economic Issues

Figure A.9: Candidate positions and congressional district composition

Notes: As in Figure 6, each panel shows the relationship between candidate positions and district demographics. Each dot represents 5% of the distribution and shows the average position of candidates, separately for Democrats and Republicans.



(a) Cultural Positions

(b) Economic Positions

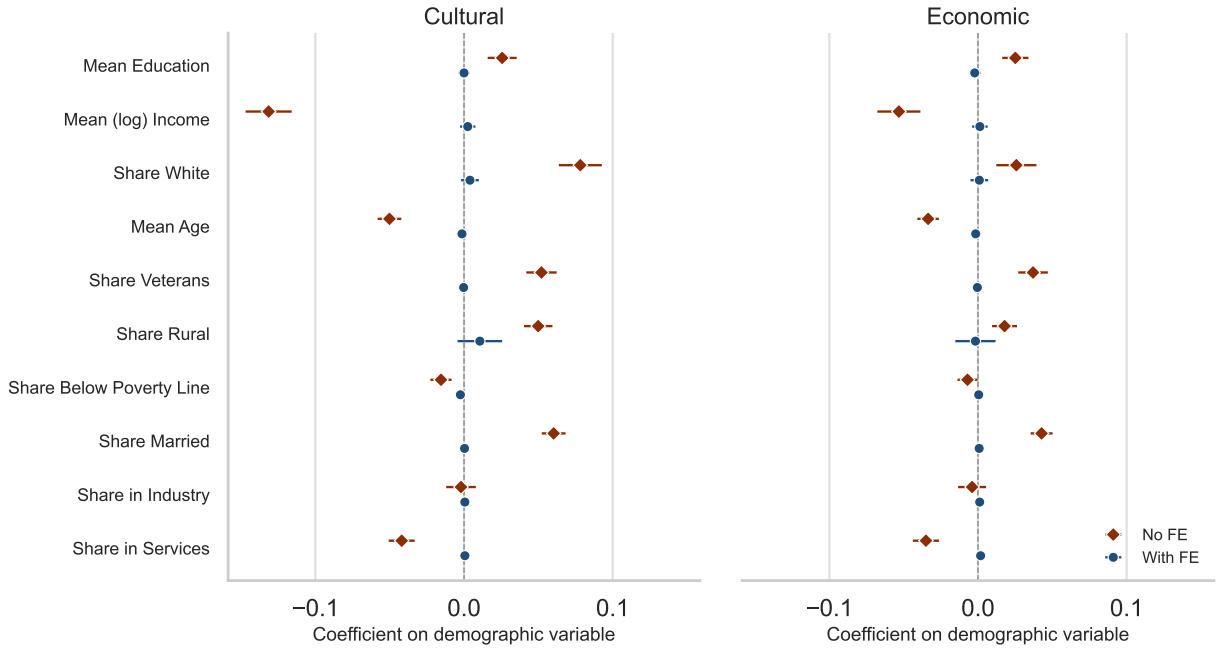
Figure A.10: Evolution of candidate positions on economic and cultural questions, separately for election winners and losers

Notes: As in Figure 5, this figure shows the evolution of the average position of candidates in each party on each dimension, separately for Congressional candidates who have won and those who have lost the election.

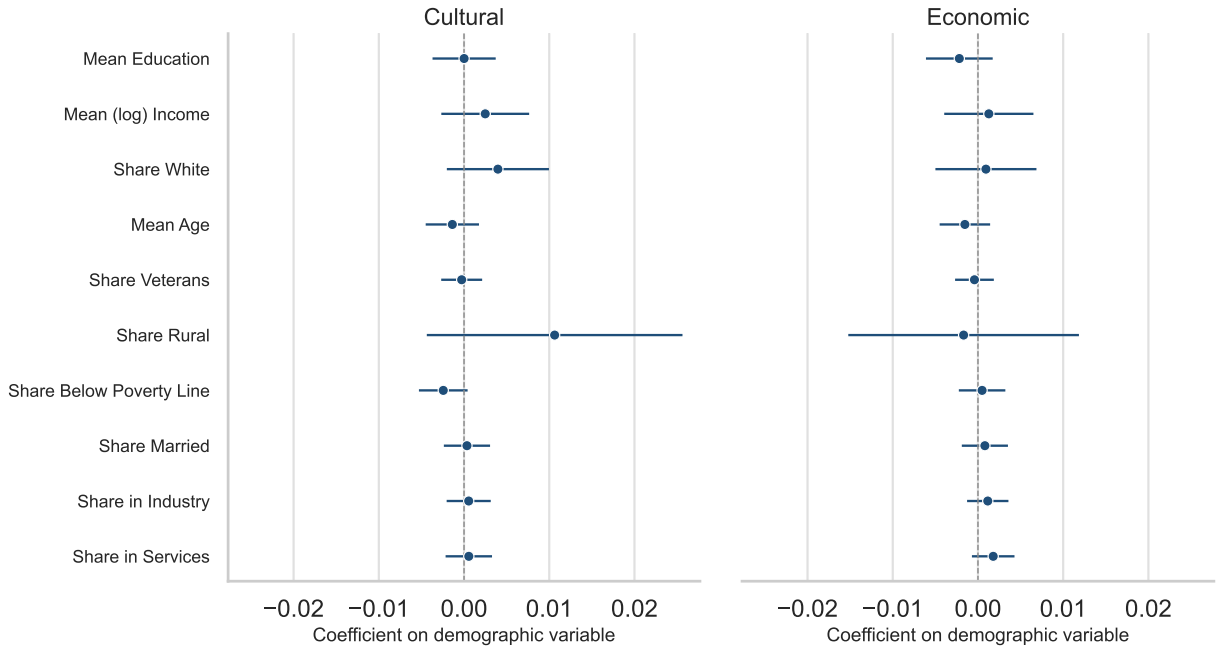
Parameter	True value	RC Logit Blockgroup Bias	RC Logit Blockgroup MSE	Logit Blockgroup Bias	Logit Blockgroup MSE	RC Logit County Bias	RC Logit County MSE	Logit County Bias	Logit County MSE
First-step coefficients									
β_{cult}	0.2500	-0.0182	0.0044	-0.2285	0.0523	-0.6539	15.8702	-0.2146	0.0541
β_{econ}	-0.4000	-0.0021	0.0019	0.8595	0.7389	-4.2152	54.9298	1.0381	1.0810
$\beta_{\text{edu,cult}}$	-0.0150	-0.0001	8.12e-06	0.0100	0.0001	0.0254	0.0708	0.0018	0.0068
$\beta_{W,\text{cult}}$	0.0300	0.0044	0.0002	-0.0132	0.0007	0.0539	0.8088	0.0826	0.1349
$\beta_{\text{age,cult}}$	0.0001	0.0005	8.56e-07	-0.0009	2.10e-06	0.0036	0.0022	0.0002	0.0004
$\beta_{\text{edu,econ}}$	0.0180	0.0002	4.31e-06	-0.0172	0.0003	0.2818	0.1912	-0.0136	0.0043
$\beta_{W,\text{econ}}$	0.0100	-0.0007	6.77e-05	-0.1209	0.0151	-1.4191	2.9546	-0.2311	0.1264
$\beta_{\text{age,econ}}$	0.0250	5.51e-06	4.51e-07	-0.0179	0.0003	0.0228	0.0090	-0.0137	0.0005
σ_{cult}	0.1800	-0.0099	0.0004	-0.1800	0.0324	0.1493	0.1071	-0.1800	0.0324
σ_{econ}	0.1200	-0.0058	0.0017	-0.1200	0.0144	0.1337	0.1040	-0.1200	0.0144
Second-step coefficients									
α_0	-7.0000	0.0271	0.0307	2.8160	7.9303	6.7400	171.2133	4.2383	17.9836
α_D	-0.7500	-0.0466	0.0703	1.0444	1.0913	-12.5690	475.4520	0.8282	0.7104
α_{edu}	0.1200	0.0004	4.53e-05	0.0096	9.37e-05	-0.4400	0.5847	-0.0638	0.0041
α_W	3.5000	-0.0084	0.0028	-2.0705	4.2870	1.8696	6.3047	-2.2503	5.0656
α_{age}	0.0400	-0.0007	5.60e-06	-0.0071	5.10e-05	-0.0259	0.0358	-0.0148	0.0002
$\alpha_{D,\text{edu}}$	0.2500	-0.0007	0.0001	-0.0195	0.0004	0.7592	1.6321	0.0459	0.0027
$\alpha_{D,W}$	-4.0000	0.0174	0.0053	-1.4126	1.9999	-2.8180	14.3018	-2.1993	4.9602
$\alpha_{D,\text{age}}$	0.0050	0.0013	1.29e-05	-0.0213	0.0005	0.0588	0.0817	-0.0183	0.0004

Table A.1: Monte Carlo bias and mean squared error across specifications

Notes: Bias and MSE from 216 Monte Carlo simulations across four specifications. The block-group-level random-coefficient multinomial logit corresponds to the paper's main specification. The table also reports a more parsimonious homogeneous-voter logit specification, as in Berry (1994), and the corresponding block-group and county-level versions of both approaches.



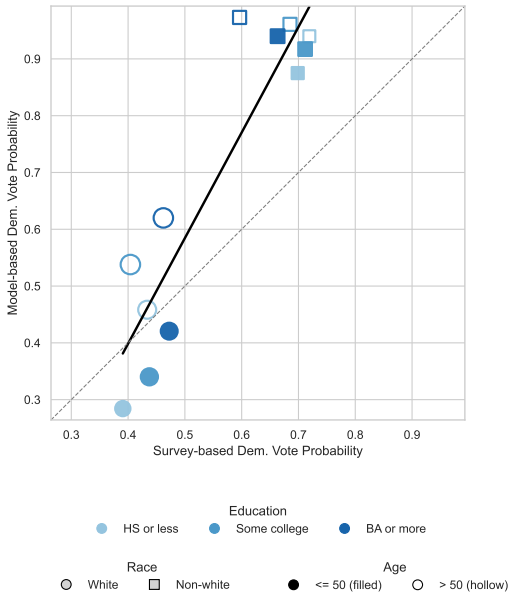
(a) With and without fixed effects



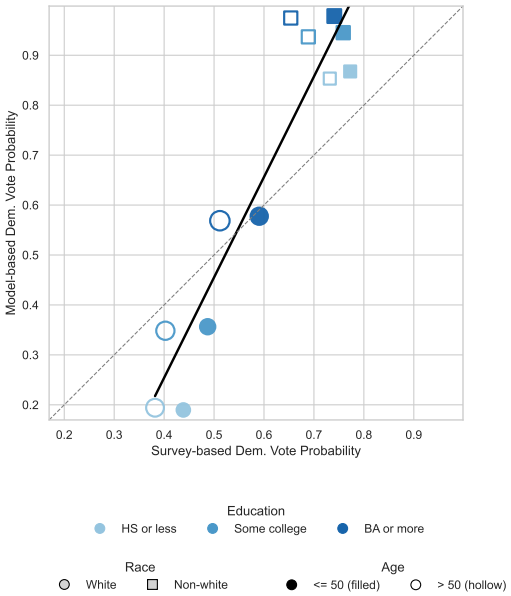
(b) With fixed effects only

Figure A.11: Balance Tests of Candidate Positions across District Boundaries.

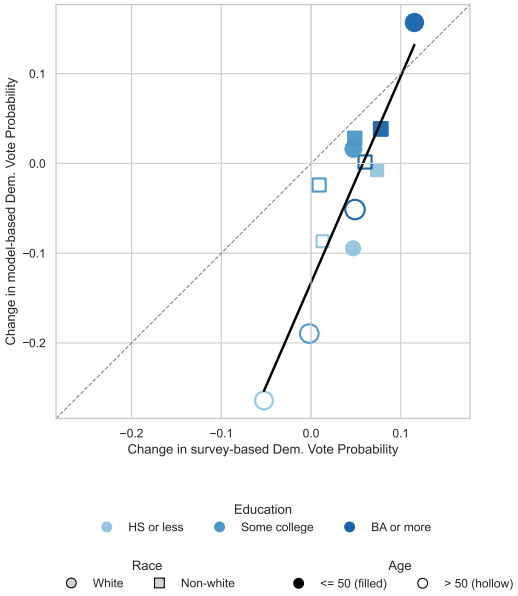
Notes: The figure reports coefficients from regressions of candidate positions on precinct-level demographic variables, using only precincts located on congressional district borders. Panel (a) reports estimates from both an OLS specification without fixed effects and a specification with precinct-by-party fixed effects and precinct-group-by-party-by-election fixed effects. Panel (b) reports only the estimates from the fixed-effects specification. Error bars denote 95% confidence intervals based on standard errors clustered at the congressional-district-by-year level. In the fixed-effects specification, the F-statistic for the joint significance of the coefficients is 1.06 ($p = 0.39$) for cultural issues and 0.50 ($p = 0.89$) for economic issues.



(a) Average, 2000–2010



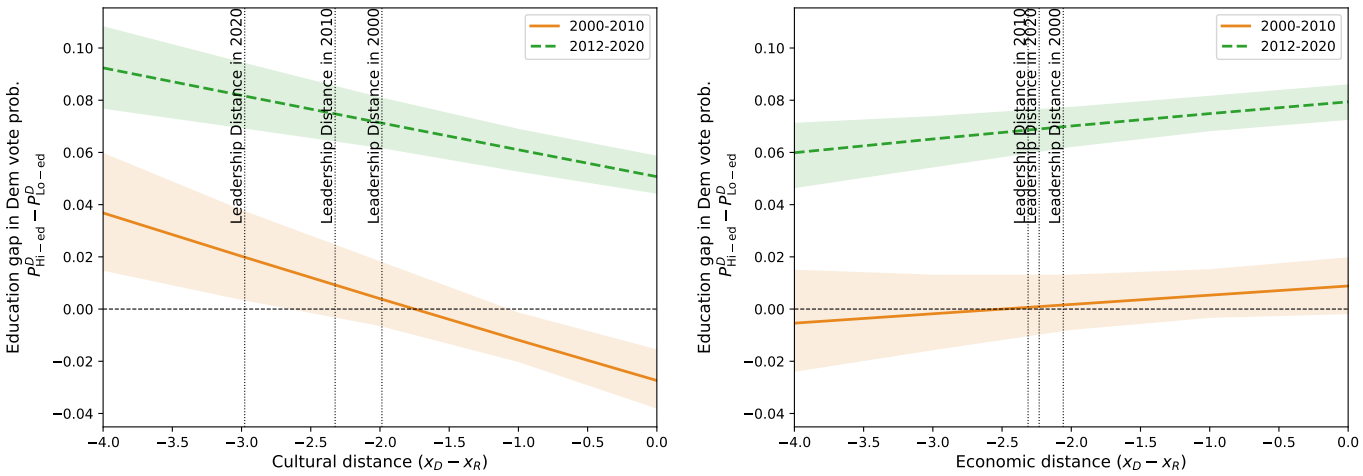
(b) Average, 2012–2020



(c) Change across periods

Figure A.12: Comparison between model-based and survey-based Democratic House vote probabilities across demographic groups

Notes: The figure compares model-based and survey-based probabilities of voting for the Democratic House candidate, conditional on voting, for 12 demographic groups defined by race, education, and age. Race is coded as White versus non-White; education is grouped into high school or less, some college, and college or more; age is grouped into 50 or below versus above 50. Panels (a) and (b) report average group-level probabilities for 2000–2010 and 2012–2020, respectively. Panel (c) reports the change in group-level probabilities between the two periods. Marker size is proportional to the number of survey respondents in each group. Colors denote education, marker shape denotes race, and filled versus hollow markers denote age group. The correlations between model-based and survey-based probabilities are 0.87 in panel (a), 0.94 in panel (b), and 0.96 in panel (c).

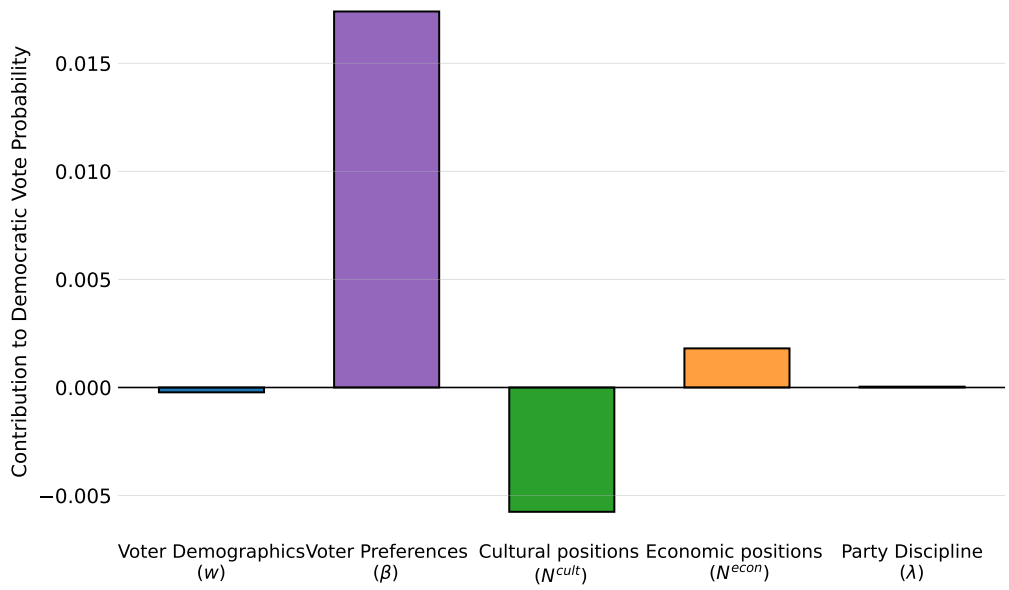


(a) Cultural Issues

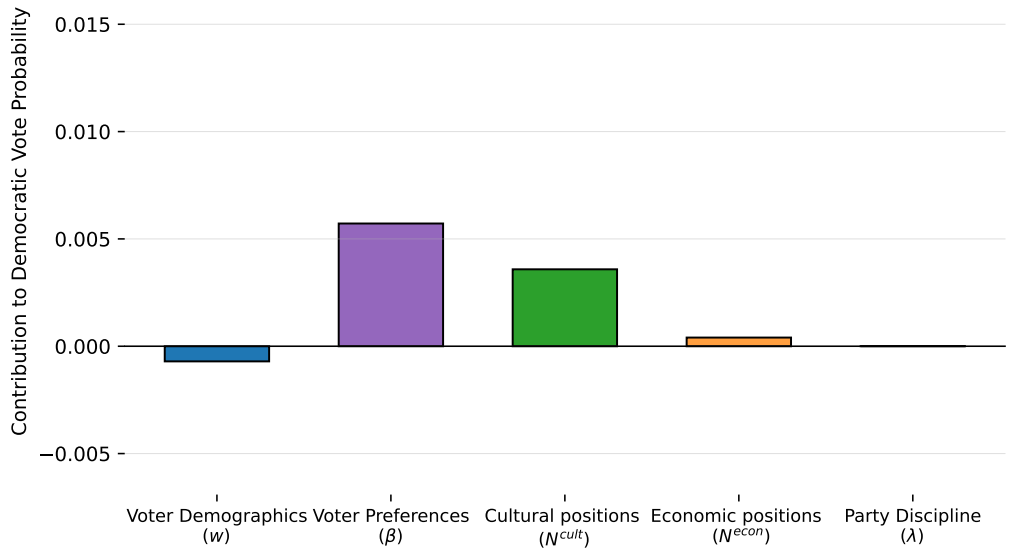
(b) Economic Issues

Figure A.13: Education Gap by Candidate Positions

Notes: The figure plots the education gap in Democratic vote probability, defined as the predicted probability that a more-educated voter supports the Democratic candidate minus the corresponding probability for a less-educated voter. Panel (a) varies the cultural distance between the Democratic and Republican candidates, holding economic distance fixed at its average 2000-2010 leadership value. Panel (b) varies economic distance, holding cultural distance fixed at its average 2000-2010 leadership value. The solid orange curve uses estimated preferences from 2000-2010, while the dashed green curve uses estimated preferences from 2012-2020. Shaded bands report bootstrap confidence intervals. Vertical dotted lines mark the observed leadership distances in 2000, 2010, and 2020. Positive values indicate that more-educated voters are more likely than less-educated voters to support the Democratic candidate; negative values indicate the reverse. Differences in levels across periods reflect changes in non-ideological parameters (α), which capture factors affecting voters' utility from supporting each party and do not vary across districts.



(a) Less-educated voters



(b) More-educated voters

Figure A.14: Equilibrium Contributions of each Factor to Changes in Vote Probability, by Education

Notes: The bars report Shapley-value contributions of each factor to the change in the predicted Democratic vote probability between the first and second periods, as described in Appendix H. The decomposition is computed in equilibrium, so each contribution incorporates both the direct effect of the corresponding factor and the indirect effect operating through endogenous candidate positioning.

C Electoral results and precinct boundaries

This section describes the methods and sources used to build the panel of electoral precincts. Since boundaries of electoral precincts change over time, I interpolate all precinct results to census block-groups, which have approximately the same population (1,875 for precincts vs. 1,600 for block-groups). Table A.2 shows the distribution of state-years included in the sample. Table A.3 shows the distribution of state-years collected and cleaned for the project but not included in the sample, either because they are not available throughout the sample or because the states have a single congressional district.

I first describe the sources used for the electoral results and the precinct shapefiles. I then explain the interpolation strategy to obtain results at the block-group level. Third, I explain how I treat absentee ballots. Finally, I describe the mechanisms implemented to check the quality of the data.

C.1 Data sources

For each election year and each state, I start by collecting all the data that was already available from previous data collection initiatives. Specifically, I collect the data available from the following six initiatives: the Harvard Election Data Archive (Ansolabehere et al., 2014, 2018), data from the MIT Election Data and Science Lab (Baltz et al., 2022), data from the Voting and Election Science Team at the University of Florida, data from the *OpenElections* initiative, data from the *Redistricting Data Hub*, and data from the *Federal Elections Project*. I also collect the U.S. Census Voting Districts (VTD) boundary shapefiles for 2000, 2010, and 2020.

If the results and/or the boundaries are missing from these sources or if the matching rate was too low, I collect the results and the boundaries myself from Secretaries of State and county election officials. I provide below the list of alternative sources. The list of state-years included in the final dataset is provided in Appendix Table A.2.

Since most of the precinct-level election results and precinct shapefiles do not come from the same sources, I match precinct-level results to their shapefiles using fuzzy-string matching within county, keeping the most likely pairs that have a normalized Levenshtein similarity above 0.7. Also, since I do not have shapefiles for every single year in the period for each state, I match the precinct-level election results with their closest shapefiles.

- For Arizona, data from 2000 was collected county by county, data for 2010 and 2014 come from the Arizona Secretary of State.
- For Arkansas, precinct shapefiles come from the Arkansas GIS Office.
- For California, all the election results and precinct boundaries come from the Statewide Database at the University of California, Berkeley.
- For Colorado, 2000 and 2002 election results are from the Colorado Secretary of State.

- For Connecticut, precinct shapefiles for 2016 come from the Metric Geometry and Gerrymandering Group
- For Florida, 2002 and 2004 results were collected county by county, and election results from 2012 onward come from the state-level Division of Elections.
- For Georgia, 2006 and 2008 results were received from the Georgia Assembly, 2012 election results come from the Secretary of State.
- For Indiana, 2012 and 2014 results were received from the Secretary of State.
- For Kentucky, results come from the State Board of Elections.
- For Massachusetts, precinct shapefiles come from the Metric Geometry and Gerrymandering Group (MGGG).
- For Michigan, results from 2000 to 2014 come from the Department of State.
- For Minnesota, election results were obtained from the Office of the Secretary of State and precinct boundaries from the Legislative Coordinating Commission of the Minnesota legislature.
- For Missouri, 2000 election results were received from the Secretary of State.
- For Montana, 2010, 2012, and 2014 election results were received from the Secretary of State.
- For Nebraska, 2010 results were received from the Secretary of State.
- For Nevada, 2004 results were received from the Secretary of State.
- For New Jersey, 2010, 2012, and 2016 results were received from the Secretary of State.
- For New Mexico, 2004, 2006 and 2010, I digitized the electoral results from PDF reports from each county.
- For North Carolina, electoral results and precinct shapefiles were obtained from the North Carolina State Board of Elections.
- For Ohio, electoral results for 2012 come from the Department of State.
- For Virginia, precinct boundaries come from Erika Lopresti's Github
- For Washington State, King County results for 2012 and 2014 come from the county election commission, 2006 results come from the Secretary of State.
- For Wisconsin, all the results and precinct boundaries come directly from the State Legislature.

I systematically use the congressional district information included in the electoral results, if the district is not reported for each precinct, I assign them using the shapefile of congressional districts compared with the precinct boundaries.

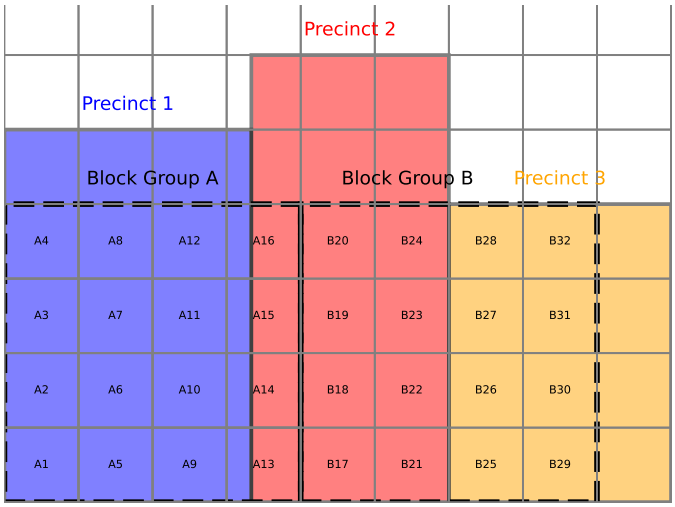


Figure A.15: Precinct votes allocation

Notes: This figure describes the strategy to allocate votes from the precinct to the block-group level. Each square of the grid shows a census block. Block-groups are shown in black and precincts are shown in blue, red, and yellow. There are about 34 census blocks per census block-group.

C.2 Panel of electoral precincts

To assign votes to each block-group, I implement the following strategy, illustrated in Figure A.15.

First, I use geospatial analysis to compare the precinct shape with each census block and create spatial crosswalks. Second, I compute the total precinct population that belongs to the block-group by using the block-level population. When a block is divided into several precincts, I assign only the share of the population that is covered by the precinct.

The total number of votes for candidate j in block-group bg can be written as:

$$Votes_{j,bg} = \sum_{p:\{p \cap bg\} \neq \emptyset} \sum_{b \in \{p \cap bg\}} Votes_{j,p} \cdot w_{b,p} \cdot \frac{Pop_b}{Pop_p}$$

where $w_{b,p}$ is the share of block b's area that falls within precinct p's boundaries, Pop_b and Pop_p are the population of the block and precinct population, respectively.

For example, consider the situation shown in Appendix Figure A.15, in which there are three precincts. In order to allocate the votes to block-group A and block-group B, I first compute the share that intersects with each block within these block-groups. Blocks are shown as grey squares of the grid. Assuming that the population of each block is equal: 80% of the population in precinct 1 belongs to block-group A (block A1 to A12 and one third of A13, A14, A15, and A16 ($w_{b,p} = \frac{1}{3}$)). Similarly, 16.7% of precinct 2 belongs to

block-group A, while 50% belongs to block-group B. 66% of precinct 3's population belongs to block-group B. The total number of votes for each candidate in block-group A, is therefore obtained by taking 80% of the votes from precinct 1 and 16.7% of the votes from precinct 2.

This strategy makes the following underlying assumptions: (1) the population is uniformly distributed within each block, (2) vote shares for each candidate are uniformly distributed within each precinct. Importantly, I do not need to assume that the population is uniformly distributed within precincts or within block-groups across blocks.

C.3 Absentee Ballots

Each state and sometimes each county reports absentee ballots differently. There are three main types of reporting of absentee votes. First, some counties directly assign absentee votes to each candidate's total in each precinct, without distinction with the election-day votes. Second, some counties record absentee ballots separately for each precinct, providing a separate estimate of the number of votes for each candidate in each precinct. Third, some counties report absentee votes at a more aggregate level than the precinct level. The rule applied in the election results has been to assign absentee ballots to specific precincts when possible. For example, Michigan counties usually report absentee ballots at the ward level, which is a more aggregated level than the precinct (about 3 precincts per ward on average). I therefore allocate the absentee votes to each precinct proportionally. If the Democratic candidate obtained 40% of their in-person votes from precinct A in ward 1, I allocate 40% of the absentee ballots from ward 1 to precinct A.

C.4 Constructing precinct-groups

For each state and election year, I first identify every pair of congressional districts that shares a common boundary. I then focus on the blockgroups that lie directly along that boundary, defined as blockgroups whose polygons touch the shared district boundary. Because raw distance measures vary sharply between urban and rural areas, I use rank-based proximity measures throughout the procedure rather than raw distances. For each blockgroup, I rank all other blockgroups in the state by proximity. I exclude block-groups for which the nearest blockgroup on the opposite side of the boundary is not among the ten closest blockgroups overall.

Some blockgroups belong to more than one congressional district; in those cases, I treat each blockgroup-by-district piece as a distinct unit. If a blockgroup touches more than one district border, I assign it to the border with which it shares the longest common boundary. Within each district pair, I then build all geographically contiguous cross-border candidate groups of size 2 to 5 and select the non-overlapping combination of groups that minimizes the total rank-based proximity cost. The algorithm is also allowed to leave some blockgroups unmatched, with a fixed penalty for doing so set equal to the initial screening threshold of ten, so poorly matched blockgroups are excluded rather than forced into low-quality groups.

State	2000	2002	2004	2006	2008	2010	2012	2014	2016	2018	2020
Alabama	x	x	x	x	x	x	x	x	x	x	x
Arizona		x	x	x	x	x	x	x	x	x	x
Arkansas	x	x	x	x	x	x	x	x	x	x	x
California	x	x	x	x	x	x	x	x	x	x	x
Colorado	x		x	x	x	x	x	x	x	x	x
Connecticut	x	x	x	x	x	x	x	x	x	x	x
Florida	x	x	x	x	x	x	x	x	x	x	x
Georgia	x	x	x	x	x	x	x	x	x	x	x
Idaho	x	x	x	x	x	x	x	x	x	x	x
Iowa	x	x	x	x	x	x	x	x	x	x	x
Kansas	x	x	x	x	x	x	x		x	x	x
Maine	x	x	x	x	x	x	x	x	x	x	x
Maryland	x	x	x	x	x	x	x	x	x	x	x
Massachusetts	x	x	x	x	x	x	x	x	x	x	x
Michigan	x	x	x	x	x	x	x	x	x	x	x
Minnesota	x	x	x	x	x	x	x	x		x	x
Mississippi			x	x	x	x	x		x	x	x
Missouri	x	x	x	x	x	x		x	x	x	x
Nevada			x	x	x	x			x	x	x
New Hampshire	x	x	x	x	x	x	x	x	x	x	x
New Jersey	x	x	x		x	x	x	x	x	x	x
New Mexico	x		x	x	x	x	x	x	x	x	x
New York	x			x	x	x	x	x	x	x	x
North Carolina	x	x	x	x	x	x	x	x	x	x	x
Oklahoma	x	x	x	x	x	x	x	x	x	x	x
Ohio			x	x	x	x	x	x	x	x	x
Oregon	x	x	x	x	x	x	x	x	x	x	x
Pennsylvania	x	x	x	x	x	x	x	x	x	x	x
Rhode Island	x	x	x	x	x	x	x	x	x	x	x
South Carolina	x			x	x	x	x	x		x	x
Tennessee		x	x	x	x	x	x	x	x	x	x
Texas	x	x	x	x	x	x	x	x	x	x	x
Virginia	x	x	x	x	x	x	x	x	x	x	x
Washington				x	x	x	x	x	x	x	x
West Virginia	x				x	x	x	x	x	x	x
Wisconsin		x	x	x	x	x	x	x	x	x	x

Table A.2: States with Electoral Results Included in the Sample

State	2000	2002	2004	2006	2008	2010	2012	2014	2016	2018	2020
Alaska		x	x	x	x	x	x	x	x	x	x
Delaware		x	x	x	x	x	x		x	x	x
Hawaii				x	x	x	x	x	x	x	x
Illinois							x	x	x	x	x
Indiana							x	x	x	x	x
Kentucky									x	x	x
Louisiana	x	x	x	x	x	x	x	x	x	x	x
Montana										x	x
Nebraska					x					x	x
North Dakota			x	x	x	x	x	x	x	x	x
South Dakota				x	x	x	x	x	x	x	x
Utah									x	x	x
Vermont		x	x	x	x	x	x	x	x	x	x
Wyoming	x	x	x	x	x	x	x	x	x	x	x

Table A.3: Electoral Results for States Not Included in the Sample

Notes: States are excluded from the sample either because their coverage is too low (e.g., Illinois or Indiana), because they are at-large districts (e.g., Delaware or Wyoming) or because they do not belong to the continental U.S. (Alaska and Hawaii).

D Estimating the Full Distribution of Precinct-Level Demographics

In order to recover the distribution of voter preferences in each precinct, I need the joint distribution of voter demographics, however, only the marginal distributions are available from U.S. Census data. I therefore implement a multi-level demographic model to recover the joint distribution. I consider two types of demographics: binary (white vs. non-white) and continuous (education, income and age). The objective is to recover the joint distribution of the three continuous variables for both white and non-white voters.

Census data from IPUMS (Manson et al., 2021) provide count variables that give the total number of people in some education, income, and age brackets, by race. These data allow me to obtain the average and the variance of each of these variables by race. Note that the estimated variance is probably underestimated since the census data only provides brackets. There are two additional challenges. First, while IPUMS provides the marginal distributions of education by race, income by race, and age by race from block-group-level data, the data do not contain the joint distribution along these three dimensions. I therefore use PUMA-level individual data from the ACS from IPUMS to obtain correlations between education, income, and age for each year. Second, since the marginal distributions by race for education and income for 2010 and 2020 are only available at the tract level, I use tract-level marginal distribution by race and block-group level distribution for the whole population to extend each block-group's 2000 marginal distribution by race.

The overall approach is described below in more detail:

In each block-group, for each decennial year, I need to estimate four matrices:

$$\boldsymbol{\mu^{white}} = (\mu^{white}_{edu}, \mu^{white}_{income}, \mu^{white}_{age}) \tag{15}$$

$$\boldsymbol{\mu^{NW}} = (\mu^{NW}_{edu}, \mu^{NW}_{income}, \mu^{NW}_{age}) \tag{16}$$

$$\boldsymbol{\sigma^{white}} = \begin{pmatrix} \sigma^{white}_{edu} & \sigma^{white}_{edu,income} & \sigma^{white}_{edu,age} \\ \sigma^{white}_{edu,income} & \sigma^{white}_{income,income} & \sigma^{white}_{income,age} \\ \sigma^{white}_{edu,age} & \sigma^{white}_{income,age} & \sigma^{white}_{age,age} \end{pmatrix} \tag{17}$$

$$\boldsymbol{\sigma^{NW}} = \begin{pmatrix} \sigma^{NW}_{edu} & \sigma^{NW}_{edu,income} & \sigma^{NW}_{edu,age} \\ \sigma^{NW}_{edu,income} & \sigma^{NW}_{income,income} & \sigma^{NW}_{income,age} \\ \sigma^{NW}_{edu,age} & \sigma^{NW}_{income,age} & \sigma^{NW}_{age,age} \end{pmatrix} \tag{18}$$

1. I use census individual-level data to get the covariance of the continuous demographics by white/non-white by PUMA (about 100 census block-groups by PUMA). This gives me the non-diagonal terms of $\boldsymbol{\sigma^{W}}$ and $\boldsymbol{\sigma^{NW}}$
2. $\boldsymbol{\mu^{white}}$ and $\boldsymbol{\mu^{NW}}$ are fully observed in 2000 at the block-group level but only partially in 2010 and

2020, for which μ_{edu}^{white} and μ_{edu}^{NW} are only reported at the tract-level.

3. For μ_{edu}^{white} and μ_{edu}^{NW} in 2010 and 2020, I estimate them by using a combination of the overall education distribution in the block-group (BG) in these years, the initial distribution of education by race in 2000 and the distribution of education by race in 2010 at the tract-level. I recover $\mu_{edu,2010,BG}^{white}$ by making the following assumption:

$$\frac{\frac{\mu_{edu,2000,BG}^{white}}{\mu_{edu,2000,BG}^{all}}}{\frac{\mu_{edu,2000,\text{tract}}^{white}}{\mu_{edu,2000,\text{tract}}^{all}}} = \frac{\frac{\mu_{edu,2010,BG}^{white}}{\mu_{edu,2010,BG}^{all}}}{\frac{\mu_{edu,2010,\text{tract}}^{white}}{\mu_{edu,2010,\text{tract}}^{all}}} \tag{19}$$

Everything is observed except $\mu_{edu,2010,BG}^{white}$. The underlying hypothesis is that the ratio of education of white to non-white in the block-group versus in the tract has stayed constant between 2000 and 2010. I recover the distributions for non-white and for 2020 in the same way.

4. The diagonal terms of $\sigma^{\mathbf{NW}}$ and $\sigma^{\mathbf{white}}$ are only observed in 2000. In the other years I just observe the diagonal terms of an aggregated matrix for all races. I therefore use the same strategy as above:

$$\frac{\frac{\sigma_{edu,2000,BG}^{white}}{\sigma_{edu,2000,BG}^{all}}}{\frac{\sigma_{edu,2000,\text{tract}}^{white}}{\sigma_{edu,2000,\text{tract}}^{all}}} = \frac{\frac{\sigma_{edu,2010,BG}^{white}}{\sigma_{edu,2010,BG}^{all}}}{\frac{\sigma_{edu,2010,\text{tract}}^{white}}{\sigma_{edu,2010,\text{tract}}^{all}}} \tag{20}$$

I recover the distributions for non-white, for 2020, and for age and income in the same way.

	Average	Overall std. deviation	Within precinct std. deviation	Within district std. deviation
Years of schooling	13.45	3.56	2.87	3.21
Age	48.91	17.98	12.87	16.32
Race (white dummy)	0.74	0.44	0.27	0.37

Table A.4: Statistics of demographics distribution

Notes: The table shows descriptive statistics on the distribution of education, age, and race and ethnicity at the individual level.

E Multimodal Text-and-Survey Ideal Point Model

This section describes the strategy I use to estimate candidate positions on multiple dimensions. I first outline the survey-based ideal point model, then describe the prediction procedure using website data, and finally explain how I aggregate both measures and report descriptive statistics.

The survey includes two main types of questions: questions asking whether a given candidate supports a policy (binary answer, such as: "Strengthening the regulation and enforcement of the Clean Water Act") or questions asking about the desired level of spending or taxes on a specific dimension, such as "Do you think inheritance taxes should be greatly increased/slightly increased/maintain status/slightly decreased/eliminated".

For each election, I recover the latent position x_{jdkt} of candidate j in district d on dimension k at election t. For expositional clarity, I suppress the district index below and write x_{jkt}.

For each candidate j, I observe responses to Q survey questions indexed by q and period t. Let y_{jqt} denote candidate j's response to item q in period t. Most items are binary, $y_{jqt} \in \{0, 1\}$, and I model them using a two-parameter logistic (2PL) item-response function. A small subset of items have ordered response categories and are modeled using a graded response model (GRM).

Binary items (2PL) For a binary item q, conditional on the latent position x_{jkt}, I model the response probability using a standard two-parameter logistic specification:

$$\Pr\big(y_{jqt} = 1 \mid x_{jkt}, \alpha_q, \beta_q\big) = \frac{1}{1 + \exp\big[-\alpha_q\big(x_{jkt} - \beta_q\big)\big]} \tag{21}$$

Here, β_q is the item cut-point (the ideological location at which a "yes" response has probability one-half), and $\alpha_q > 0$ is the discrimination parameter, capturing how sharply the response probability changes with the latent position.

Ordered items (GRM) For an ordered item q with C_q response categories, $y_{jqt} \in \{0, 1, \ldots, C_q - 1\}$, I model responses using a graded response specification. Conditional on the latent position x_{jkt}, the cumulative response probabilities are modeled as:

$$\Pr\big(y_{jqt} \geq c \mid x_{jkt}, \alpha_q, \delta_q\big) = \frac{1}{1 + \exp\big[-\alpha_q\big(x_{jkt} - \delta_{qc}\big)\big]}, \qquad c = 1, \ldots, C_q - 1. \tag{22}$$

Here, δ_{qc} are ordered cut-points with $\delta_{q1} < \cdots < \delta_{q,C_q-1}$, which partition the latent ideological scale into adjacent response regions, and $\alpha_q > 0$ is the discrimination parameter, governing how sharply the probability of higher-category responses varies with the latent position.

Category probabilities are obtained from differences of adjacent cumulative probabilities:

$$\Pr(y_{jqt} = 0 \mid x_{jkt}, \alpha_q, \delta_q) = 1 - \Pr(y_{jqt} \geq 1 \mid \cdot), \tag{23}$$

$$\Pr(y_{jqt} = c \mid x_{jkt}, \alpha_q, \delta_q) = \Pr(y_{jqt} \geq c \mid \cdot) - \Pr(y_{jqt} \geq c+1 \mid \cdot), \qquad c = 1, \ldots, C_q - 2, \tag{24}$$

$$\Pr(y_{jqt} = C_q - 1 \mid x_{jkt}, \alpha_q, \delta_q) = \Pr(y_{jqt} \geq C_q - 1 \mid \cdot) \tag{25}$$

Latent positions evolve as an AR(1) process,

$$x_{jkt} = \rho_k x_{jk,t-1} + \sigma_k \eta_{jkt}, \qquad \eta_{jkt} \sim \mathcal{N}(0,1) \tag{26}$$

I parameterize persistence in terms of a half-life h_k such that shocks decay by 50% after h_k periods:

$$\rho_k = 2^{-1/h_k} \tag{27}$$

Rather than fixing h_k, I estimate it from the data by placing a prior on its log scale,

$$\log h_k \sim \mathcal{N}(\log h_{0k}, \tau_h^2) \tag{28}$$

In the baseline specification, the prior is centered on one election cycle, so ρ_k is centered on $\frac{1}{2}$. To fix the scale of the latent dimension, I impose stationarity with unit variance,

$$\sigma_k = \sqrt{1 - \rho_k^2} \tag{29}$$

and set the initial condition

$$x_{jk0} \sim \mathcal{N}(0,1) \tag{30}$$

For binary items, I place the priors $\beta_q \sim \mathcal{N}(0,1)$ and $\alpha_q \sim \text{LogNormal}(0, 0.25)$. For graded items, I use $\alpha_q \sim \text{LogNormal}(0, 0.25)$ and parameterize ordered cut-points via a free baseline $\delta_{q1} \sim \mathcal{N}(0,1)$ and independent positive increments $\Delta_{qc} \sim \text{HalfNormal}(1)$, defining

$$\delta_{qc} = \delta_{q,c-1} + \Delta_{qc}, \qquad c = 2, \ldots, C_q - 1,$$

so that $\delta_{q1} < \cdots < \delta_{q,C_q-1}$.

All model parameters are estimated with Bayesian Markov chain Monte Carlo implemented in PyMC. Responses that are not observed are treated as missing-at-random and excluded from the likelihood. For each candidate-period ideal point, I take the posterior mean as the estimate and use the posterior standard deviation of the MCMC draws as its standard error.

Appendix Table A.5 summarizes the number of raw questions by topic and their classification into issues. I exclude questions that mention a specific country which are likely to be context specific (e.g., "Should the United States increase financial support to Afghanistan?", "Should the U.S. support the creation of a Palestinian state"). I also exclude four questions that were misinterpreted by candidates: e.g., "Aid should be granted to countries when it is in the security interests of the U. S." was interpreted by some candidates as "only in these cases", which leads to some internal contradictions in candidates' answers. I also exclude the question: "Should trade agreements include provisions to address environmental concerns and to protect workers' rights?" since it could be classified as either "trade" or "environment" and therefore either cultural or economic. I also exclude the question: "Prohibit the dilation and extraction procedure.", asked in 2008 and 2010, which gives very inconsistent responses by the same candidates from one year to the next.

Topic	Dimension	Number of distinct raw questions
Abortion	Cultural	7
Crime	Cultural	11
Education	Cultural	7
Environment	Cultural	17
Labor and Employment	Economic	9
Gun regulations	Cultural	8
International Trade	Economic	6
Campaign Finance	Cultural	4
Immigration	Cultural	8
Diversity Questions	Cultural	3
Health Care	Economic	9
Taxes and Spending	Economic	19
Security and International Policy	Cultural	7
Social Security	Economic	13
Welfare	Economic	4

Table A.5: Summary of topic classifications

Once I obtain an ideal point and a standard error for each candidate that has answered the survey, I train and apply a machine learning regressor (Extreme Gradient Boosting) using features extracted from their website. I scrape the website in the following way: I collect all the text available on the first page, all the text available on any page listed on the first page and iterate once again by taking all the text available on any page listed on these second pages. I only collect text that is on the website (i.e., I do not scrape external websites even if they are referenced on the candidate's website). Starting from the raw text data scraped from candidate websites, I clean the text in regular ways by removing all words that are a consequence of the data being displayed on a website (e.g., "contact me", "send an email", "access photos", etc.), removing names of candidates, state names and region names. I then construct unigrams, bigrams and trigrams (sequences of 1, 2 or 3 words) and I remove words that are too infrequent

(used in less than 0.05% of the documents), which gives me a large matrix of occurrences of tokens for each candidate. In addition, I also compute document embeddings (Dai et al., 2015) which give a vector representation of each document.

The mean squared error (MSE) for the economic prediction is 0.17 (R-squared of the prediction is 0.91) and 0.21 for the cultural dimension (R-squared of 0.89). For candidates for whom I only have the website (7%) of the sample, I only assign the website ideal point, for those with only the survey (32%), I only assign the survey ideal points and for all those with both survey and website ideal points (49%), I take a weighted average of the two measures using the following formula:

$$x_{jkt} = \omega_{jkt} x_{jkt}^{\text{survey}} + (1 - \omega_{jkt}) x_{jkt}^{\text{website}}, \tag{31}$$

with $\omega_{jkt} = \frac{MSE_k^{website}}{se(x_{jkt}^{\text{survey}})^2 + MSE_k^{website}}$.

Figure A.16 shows the correlation between the survey-based and website-based measures.



(a) All Sample



(b) Democrats only



(c) Republicans only

Figure A.16: Comparison between survey-based and website-based measures

Notes: The figure shows the pairwise correlation between survey-based and website-based measures. The first panel looks at the overall correlation. The second and third panels show the within-party correlations for Democrats and Republicans, respectively.

F Additional Demand Results

This section explores the robustness of the demand results in two dimensions: using a simple specification following Berry (1994) and controlling for additional candidate observable characteristics that could correlate with candidate ideology.

F.1 Homogeneous-Voter Logit Specification

Assuming that voters are homogeneous within precincts enables the estimation of a simpler model, following Berry (1994). Specifically, one can write:

$$\log\left(\frac{S_{jpt}}{S_{0pt}}\right) = \alpha_{jt} + \mathbf{w}'_{pt}\boldsymbol{\alpha}^{\mathbf{w}}_{jt} + \sum_{k}\left(\beta_{kt}x_{jdkt} + \mathbf{w}'_{pt}\boldsymbol{\beta}^{\mathbf{w}}_{kt}x_{jdkt}\right) + \xi_{jpt}, \tag{32}$$

where S_{jpt} denotes observed vote shares. Here, α_{jt} is the average propensity to vote for a candidate of party j at time t, $\boldsymbol{\alpha}^{\mathbf{w}}_{jt}$ represents the effect of precinct-level voter demographics $\mathbf{w}_{pt}$ on baseline partisan voting, and x_{jdkt} is party j's candidate position in district d on dimension k, $\beta^{\mathbf{w}}_{kt}$ captures voter heterogeneity along dimension w for dimension k.

Table A.6 reports the coefficients from Equation (32) estimated by OLS, with precinct group by party by election fixed effects and precinct by party fixed effects. Results are quite similar to the coefficients estimated by the main specification.

	Outcome: ln(sj/s0)	
	(1)	(2)
	2000-2010	2012-2020
Cultural Position	0.004	-0.009
	(0.017)	(0.017)
Economic Position	0.020	-0.021
	(0.018)	(0.019)
Dem=1 × Av. Edu	-0.052*	0.056*
	(0.031)	(0.031)
Dem=1 × Shr. White	-0.008	-0.082**
	(0.038)	(0.036)
Dem=1 × Av. Age	-0.032	0.025
	(0.027)	(0.023)
Av. Edu	0.064***	0.054***
	(0.022)	(0.019)
Shr. White	0.137***	0.203***
	(0.030)	(0.024)
Av. Age	0.080***	0.078***
	(0.017)	(0.013)
Cultural Position × Av. Edu	-0.028*	-0.023*
	(0.014)	(0.013)
Cultural Position × Shr. White	-0.008	-0.006
	(0.017)	(0.014)
Cultural Position × Av. Age	-0.006	0.014
	(0.012)	(0.009)
Economic Position × Av. Edu	0.003	0.056***
	(0.014)	(0.014)
Economic Position × Shr. White	0.011	-0.034**
	(0.017)	(0.015)
Economic Position × Av. Age	-0.010	0.001
	(0.012)	(0.010)
Precinct-pair x Party x Year FE	X	X
Precinct x Party FE	X	X
Observations	179,124	205,104

Table A.6: Estimation of Voter Preferences with Homogeneous Voters

Notes: This table shows the coefficients from Equation (32): a regression of candidates' log odds ratio on interactions of precinct-level demographics and candidate positions, by period. Demographic variables are normalized to have unit variance. Each column controls for precinct-by-party fixed effects and precinct-pair-by-party-by-election fixed effects. Standard errors clustered by congressional district by year are reported in parentheses.

F.2 Additional candidate observable characteristics

While the identification strategy handles any unobservable voter taste shocks that would span across the congressional district border, one might wonder whether voter preferences for ideology might instead capture preferences of voters for candidate characteristics that would vary with ideology. This section tests the sensitivity of the demand results to the addition of three candidate observable characteristics: incumbency, gender, and race. I code candidate gender using name-classification algorithms. I measure candidate race and ethnicity using Bouton et al. (2022) for House elections after 2006, the ANES contextual files for 2002 and 2004, and the official House History minority-member tables for House winners in 2000. I exclude elections for which the gender or race or ethnicity of either candidate cannot be determined. While other dimensions such as age likely play a role as well, they are rarely observed for non-incumbents. For each additional observable characteristic, I recover heterogeneity across demographic variables by estimating the following voter random utility model:

$$u_{ijpt} = \sum_k \beta_{ikt} x_{jdkt} + \alpha_{ijt} + \gamma^1_{it} Inc_{jdt} + \gamma^2_{it} Fem_{jdt} + \gamma^3_{it} NonWhite_{jdt} + \xi_{jpt} + \epsilon_{ijpt} \tag{33}$$

where Inc_{jdt} is a dummy equal to one if candidate j is the incumbent, Fem_{jdt} is a dummy equal to one if candidate j is a woman, $NonWhite_{jdt}$ is a dummy equal to one if candidate j is non-white, and all the other parameters as defined in equation (6).

I recover preferences under specification 33 using the random-coefficient multinomial logit described in section 4. The coefficients on ideology are quite similar to the estimates from the main specification. The coefficients in the second period are smaller in magnitude but the relative effect of cultural versus economic issues is preserved.

Coefficients	Parameters	Estimates	Standard Errors
First Period: 2000-2010			
Cult	β_{cult}	0.3830	(0.0913)
Econ	β_{econ}	-0.1436	(0.1094)
Incumbent	β_{inc}	0.1125	(0.0884)
Female	β_{female}	-0.1381	(0.0907)
Candidate nonwhite	β_{nonwhite}	0.1107	(0.0931)
Baseline Yrs. Schooling	α_{edu}	0.0230	(0.0099)
Baseline White	α_W	0.7046	(0.0859)
Baseline Age	α_{age}	0.0129	(0.0024)
Cult × Yrs. Schooling	$\beta_{\text{edu,cult}}$	-0.0275	(0.0060)
Cult × White	$\beta_{W,\text{cult}}$	-0.0484	(0.0373)
Cult × Age	$\beta_{\text{age,cult}}$	0.0006	(0.0016)
Econ × Yrs. Schooling	$\beta_{\text{edu,econ}}$	0.0183	(0.0063)
Econ × White	$\beta_{W,\text{econ}}$	-0.0550	(0.0362)
Econ × Age	$\beta_{\text{age,econ}}$	-0.0014	(0.0018)
Incumbent × Yrs. Schooling	$\beta_{\text{inc,edu}}$	-0.0107	(0.0054)
Incumbent × White	$\beta_{\text{inc},W}$	-0.0197	(0.0400)
Incumbent × Age	$\beta_{\text{inc,age}}$	0.0081	(0.0016)
Female × Yrs. Schooling	$\beta_{\text{female,edu}}$	-0.0006	(0.0057)
Female × White	$\beta_{\text{female},W}$	-0.0138	(0.0363)
Female × Age	$\beta_{\text{female,age}}$	0.0047	(0.0015)
Candidate nonwhite × Yrs. Schooling	$\beta_{\text{nonwhite,edu}}$	-0.0120	(0.0065)
Candidate nonwhite × White	$\beta_{\text{nonwhite},W}$	-0.1690	(0.0509)
Candidate nonwhite × Age	$\beta_{\text{nonwhite,age}}$	0.0052	(0.0019)
Dem × Yrs. Schooling	$\alpha_{D,\text{edu}}$	-0.0111	(0.0135)
Dem × White	$\alpha_{D,W}$	-0.4780	(0.0966)
Dem × Age	$\alpha_{D,\text{age}}$	-0.0145	(0.0035)
Unobserved cultural heterogeneity	$\sigma^{\beta}_{\text{cult}}$	0.0039	(0.0122)
Unobserved economic heterogeneity	$\sigma^{\beta}_{\text{econ}}$	0.0346	(0.0103)
Second Period: 2012-2020			
Cult	β_{cult}	0.0621	(0.0919)
Econ	β_{econ}	-0.1471	(0.0842)
Incumbent	β_{inc}	-0.3513	(0.0879)
Female	β_{female}	0.2685	(0.0942)
Candidate nonwhite	β_{nonwhite}	0.0497	(0.0851)
Baseline Yrs. Schooling	α_{edu}	0.0120	(0.0100)
Baseline White	α_W	1.0078	(0.0761)
Baseline Age	α_{age}	0.0059	(0.0026)
Cult × Yrs. Schooling	$\beta_{\text{edu,cult}}$	-0.0075	(0.0055)
Cult × White	$\beta_{W,\text{cult}}$	-0.1084	(0.0378)
Cult × Age	$\beta_{\text{age,cult}}$	0.0033	(0.0014)
Econ × Yrs. Schooling	$\beta_{\text{edu,econ}}$	0.0102	(0.0048)
Econ × White	$\beta_{W,\text{econ}}$	-0.0413	(0.0325)
Econ × Age	$\beta_{\text{age,econ}}$	0.0001	(0.0015)
Incumbent × Yrs. Schooling	$\beta_{\text{inc,edu}}$	-0.0024	(0.0047)
Incumbent × White	$\beta_{\text{inc},W}$	-0.0408	(0.0298)
Incumbent × Age	$\beta_{\text{inc,age}}$	0.0149	(0.0017)
Female × Yrs. Schooling	$\beta_{\text{female,edu}}$	-0.0040	(0.0051)
Female × White	$\beta_{\text{female},W}$	-0.0536	(0.0414)
Female × Age	$\beta_{\text{female,age}}$	-0.0041	(0.0014)
Candidate nonwhite × Yrs. Schooling	$\beta_{\text{nonwhite,edu}}$	-0.0015	(0.0056)
Candidate nonwhite × White	$\beta_{\text{nonwhite},W}$	0.0102	(0.0364)
Candidate nonwhite × Age	$\beta_{\text{nonwhite,age}}$	-0.0001	(0.0015)
Dem × Yrs. Schooling	$\alpha_{D,\text{edu}}$	0.0140	(0.0126)
Dem × White	$\alpha_{D,W}$	-0.3639	(0.0985)
Dem × Age	$\alpha_{D,\text{age}}$	0.0084	(0.0035)
Unobserved cultural heterogeneity	$\sigma^{\beta}_{\text{cult}}$	0.0102	(0.0125)
Unobserved economic heterogeneity	$\sigma^{\beta}_{\text{econ}}$	0.0108	(0.0103)

Table A.7: Robustness to the Inclusion of Candidate Observable Characteristics

Notes: The table provides estimates from equation 33, estimated using the random-coefficient multinomial logit, including preference heterogeneity along candidate observable characteristics (incumbency, gender, and race and ethnicity).

G Alternative supply specification

Section 5 estimates a supply model in which the probability of winning is equal to the vote share plus a uniformly distributed shock. This assumption makes the probability of winning a linear function of the vote share. This section tests the robustness of the supply estimates to that assumption by instead allowing the shock to follow a logistic distribution. This breaks the linear relationship between the probability of winning and vote share and makes uncertainty more concentrated in races close to 50%.

I write each candidate's objective function as Π_{jt}, which depends on the candidate's vote share and the distance between their chosen position and the party leadership.

$$\Pi_{jt}(\mathbf{x_{jt}}) = \lambda_{jt} \underbrace{P_j(\mathbf{x_{jt}}, \mathbf{x_{-jt}})}_{\text{probability of winning}} - \underbrace{||\mathbf{x_{jt}} - \mathbf{N_{jt}}||^2}_{\text{distance from national party platform}} \tag{34}$$

where the parameter λ_{jt} captures the weight candidates give to electoral incentives when choosing their position.

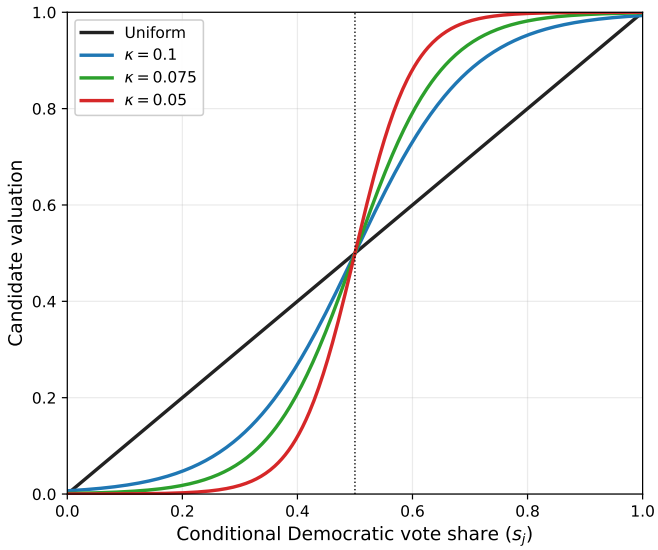
There is an aggregate shock ζ_j that creates some uncertainty around candidates' probability of winning.

If, instead of a uniform distribution as used in the main paper, ζ follows a Logistic distribution, which allows the probability of winning to be a non-linear function of the vote share, the probability of winning can be re-written as:

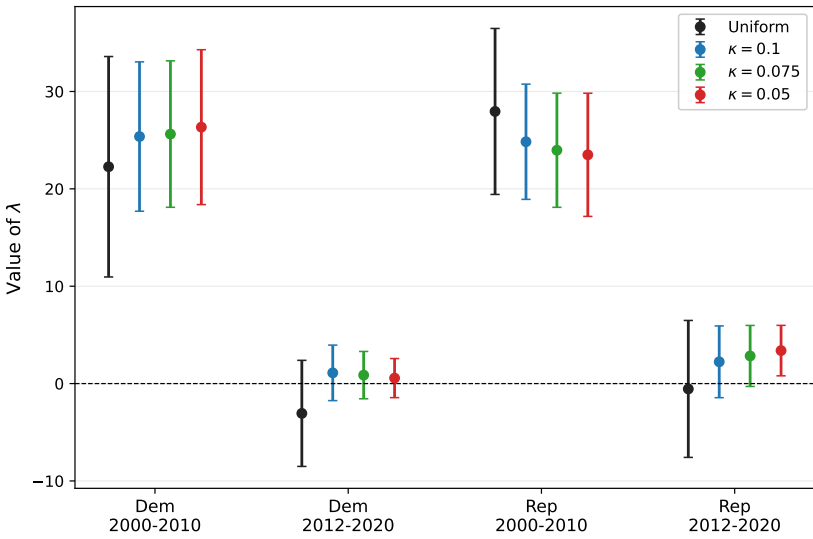
$$\begin{aligned} P_j(\mathbf{x_{jt}}, \mathbf{x_{-jt}}) &= \mathbb{P}(s_j(\mathbf{x_{jt}}, \mathbf{x_{-jt}}) + \zeta_j \geq 0.5) \\ &= 1 - \mathbf{P}(\zeta_j \leq 0.5 - s_j(\mathbf{x_{jt}}, \mathbf{x_{-jt}})) \\ &= \frac{1}{1 + \exp\big(-\frac{s_j - 0.5}{\kappa}\big)} \end{aligned}$$

where κ is the scale of the Logistic distribution.

Figure A.17 reports different measures of the estimate of party discipline λ for different values of the scale of the Logistic distribution (κ). The findings are quantitatively similar for each value of κ and the ratio between λ for a given value of κ is almost identical. Overall, lower values of κ lead mostly to larger estimates of party discipline since, for most candidates, the slope of candidates' valuation becomes steeper.



(a) Candidate valuation for different κ



(b) Estimated λ

Figure A.17: Estimated λ for different values of κ.

Notes: The figure shows estimated levels of party discipline when relaxing the linearity assumption of the candidate valuation of their vote share. I re-estimate λ where the voting aggregate shock follows a Logistic distribution instead of a uniform distribution. The figure reports different estimates of λ for different scale parameters of the Logistic distribution κ with location 0.5. Panel (a) shows different shapes of candidates' valuation for different values of κ. Panel (b) shows the corresponding estimates of party discipline.

H Details on Realignment Counterfactuals

This section details the strategy to measure the contributions of each demand-side and supply-side factor to the change in equilibrium outcomes: candidate positioning and voting behavior. Each counterfactual scenario is characterized by a set of parameters that give a unique equilibrium defined by $(\mathbf{x}^*, \mathbf{s}^*) = f\left(F_{pt}\left(\mathbf{w}\right), \boldsymbol{\beta_t}, \mathbf{N_t}, \boldsymbol{\lambda_t} \middle| \mathbf{m_t}, \boldsymbol{\alpha_t}, \boldsymbol{\xi_t}, \boldsymbol{\eta_t}\right)$ where $\mathbf{x}^*, \mathbf{s}^*$ are vectors of candidate positions and vote probabilities, $F_{pt}(\mathbf{w})$ is the distribution of demographics in each precinct p, $\boldsymbol{\beta_t}$ captures voter ideological preferences, $\mathbf{N_t}$ denotes national party leaders' positions, $\boldsymbol{\lambda_t}$ measures the relative importance of electoral incentives, with lower values corresponding to stronger party discipline, $\mathbf{m_t} = (m_{p,d})_{\forall p,d}$ is a redistricting scenario, that is, a map of precincts to districts, $\boldsymbol{\alpha_t}$ accounts for voter partisan preferences, $\boldsymbol{\xi_t}$ is a vector of precinct-level taste shocks, and $\boldsymbol{\eta_t}$ is a vector of candidate-positioning residuals.

I consider five factors separately: (a) Changes in the demographic distribution ($F_t\left(\mathbf{w}\right)$), (b) changes in voter preferences ($\boldsymbol{\beta_t}$), (c) changes in the party leadership positions ($\mathbf{N_t}$), with (c1) changes in cultural positions and (c2) changes in economic positions, and (d) changes in the strength of party discipline ($\boldsymbol{\lambda_t}$).

Each scenario is defined by a combination of factors, evaluated either at their initial or final values. With five factors of interest plus two nuisance factors, this results in $2^7 = 128$ counterfactual scenarios.[18] To assess the contribution of each factor, I compare its effect on the equilibrium outcomes, holding all other factors at various permutations of their initial and final values. The results are aggregated across scenarios using a Shapley value decomposition (Shapley, 1953), following the approach of Guriev et al. (2023).

For each factor $z \in \{F_p(\mathbf{w}), \boldsymbol{\beta}, \mathbf{N_{cult}}, \mathbf{N_{econ}}, \boldsymbol{\lambda}, \boldsymbol{\alpha}, \boldsymbol{m}\}$, its contribution to the change in voter i's probability of voting for the Democratic candidate $s_{ij}(\cdot)$ is given by:

$$\phi_i(z) = \frac{1}{n} \sum_{C} \binom{n-1}{|C|}^{-1} \left(s_{ij}(\cdot, z_{final}) - s_{ij}(\cdot, z_{initial})\right)$$

for each "coalition" C not including z. A coalition C here denotes the factors that are evaluated at their final rather than initial value. $n = 7$. I then average the contributions across individuals, separately for more-educated and less-educated individuals. I apply the same strategy to decompose the change in candidate positions.

[18] While I am interested only in recovering the impact of the five factors above, two further dimensions to which I do not give a structural interpretation also change over the same period: overall partisanship ($\boldsymbol{\alpha}$) and the district map ($\mathbf{m}$). Rather than fix these at an arbitrary initial or final value, I include them in the Shapley computation as well, which is why the decomposition spans seven dimensions in total. Figure 9 reports the contributions of the five focal factors only.